



INTS402 Capstone Project Final Report

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Interpreting Digital Emotion in Youth Support: Designing
a Copilot System for Singapore-Based Peer Mentors

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Acknowledgements

Ten years ago, a 15-year-old version of Wooz returned from a community service trip to India with a vague but persistent sense that I wanted to build things that could help - not just donate pencils, but contribute to deeper change by understanding the structures that made them necessary in the first place.

This vision was later sharpened by my National Service experience, where witnessing mental health incidents up close revealed the gaps between institutional support systems and human emotional needs. To those who stood beside me during that difficult period: thank you for showing me that compassion and systematic support need not be opposing forces - a lesson that directly shaped this project's approach to emotional intelligence augmentation.

As this capstone marks the final submission of my undergraduate journey at SMU, I want to acknowledge the people, places, and moments that shaped not just this project, but the broader arc of how I've come to think, work, and reflect.

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This project is a distillation of many threads. I'm grateful to the College of Integrative Studies for providing the environment where I could tie them together, however loosely, in these final pages.

And to that 15-year-old version of me, this is me, still trying.

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Post-Capstone Reflection

Executive Summary

This project developed an AI-augmented copilot system designed to assist youth well-being mentors in interpreting digital emotional expressions, addressing the growing complexity of asynchronous and multimodal communication in Singapore's mental health support landscape. The system was built in partnership with Over-The-Rainbow (OTR), a local NGO offering peer-based emotional support to youth. Grounded in a human-centered design methodology, the development process combined qualitative research, user journey mapping, iterative prototyping, and structured testing with OTR mentors to ensure contextual and practical relevance.

Findings from the project highlight several key insights. Peer mentors consistently reported a significant cognitive load when interpreting emotional signals in digital chat environments, particularly in the absence of tailored tools. The integration of AI support improved interpretation accuracy by 65 percent, but this benefit was only realized when the system was seamlessly embedded within existing workflows. Mentors emphasized the unique value of visual meta-analysis features, which allowed them to detect emotional patterns across conversations—an enhancement they viewed as particularly useful for supporting long-term relationships. Notably, basic user experience elements played a decisive role in shaping trust in the system; participants expressed that unless foundational usability was met, confidence in the AI's advanced features remained limited.

Beyond its immediate application, the project contributes to the theoretical development of digital emotional intelligence (dEI) by incorporating computational and collaborative dimensions into traditionally psychological frameworks. It also offers a working prototype that provides concrete design principles for future AI integration in youth mental health contexts. Emphasizing augmentation over automation, the system demonstrates how technology can support—rather than replace—the relational core of peer-based emotional care.

Introduction

The digital transformation of mental health support services has fundamentally altered how youth express emotional experiences. While technology has expanded accessibility of support, it has simultaneously created new challenges for human supporters who must interpret and respond to digital emotional expressions without traditional nonverbal cues. This gap between conventional emotional support frameworks and the complex reality of digital emotional expression represents a challenge for organizations like Over-The-Rainbow (OTR), a Singapore-based non-profit offering peer support to youth.

This challenge is especially salient in Singapore, where digital support services are increasingly being scaled to meet rising demand among youths navigating academic pressure, family expectations, and identity concerns. The linguistic hybridity of digital expression (e.g., Singlish, code-switching, emoji blends) and cultural stigma around emotional disclosure further complicate efforts to support help-seeking behaviors effectively in online contexts.

OTR operates OTR Listens, a digital peer support platform providing real-time chat, self-care resources, and a virtual relaxation space for youth seeking emotional support. OTR's volunteers, known as "Avies," face significant challenges in accurately interpreting emotional cues in digital conversations. During initial research with OTR supporters, many reported spending "considerably more time" interpreting digital expressions compared to face-to-face interactions, suggesting a substantial efficiency gap in the digital support workflow.

Traditional emotional intelligence approaches, which were primarily developed for face-to-face interactions, are insufficient for capturing the distinct characteristics of digital communication. These limitations become especially evident when considering the asynchronous nature of online messaging and the irregular timing patterns it introduces. Moreover, digital conversations often include complex multimodal expressions that blend emojis, images, and text, making interpretation more challenging. The absence of immediate feedback and nonverbal cues further compounds these difficulties. Additionally, individual variations in baseline communication styles and significant cultural or generational differences in how emotions are digitally expressed contribute to the interpretative complexity mentors face in online support contexts.

This project addresses these challenges by developing an evidence-based "copilot" system for OTR's well-being mentors that augments-rather than replaces-human capabilities in interpreting digital emotional expressions. The system integrates with OTR's existing platform to enhance well-being mentors' ability to assess emotional urgency and contextualize ambiguous expressions in their digital interactions with youth.

The project is guided by three central hypotheses:

1. **Augmentation Hypothesis:** A well-designed "copilot" system combining psychological frameworks with computational linguistics will enhance well-being mentors' ability to accurately interpret digital emotional expressions compared to unaided interpretations.
2. **Integration Hypothesis:** The effectiveness of such a system depends on its integration into existing support workflows rather than disrupting them with entirely new processes.
3. **Complementary Intelligence Hypothesis:** The most effective approach enhances rather than replaces human judgment by combining human empathy with AI-driven pattern recognition, providing contextual insights rather than definitive interpretations.

By integrating psychological frameworks with computational linguistics and human-computer interaction principles, this project aims to develop a system that serves as an intelligent assistant to human supporters, enhancing their capabilities while preserving the essential human element of emotional support.

Literature Review

Emotional Intelligence in Digital Contexts

Traditional emotional intelligence frameworks provide essential structure for understanding emotions but require specific adaptation for digital contexts. The RULER framework (Brackett et al., 2019) was selected for this project over alternatives like the Goleman model because it offers five operationalizable components-Recognizing, Understanding, Labeling, Expressing, and Regulating emotions-that can be systematically mapped to digital communication patterns. While RULER provides strong conceptual foundations, its development in educational contexts creates limitations for direct application to asynchronous digital support conversations where emotional expressions are fragmented across time and modalities.

Digital communications present unique interpretative challenges not adequately addressed by traditional frameworks. As Kaye et al. (2017) demonstrate, asynchronous interaction patterns create temporal ambiguity where emotional content might be separated from its context by minutes, hours, or days. This temporal separation represents a fundamental challenge that traditional emotional intelligence frameworks, developed for real-time interaction, fail to address methodologically.

Recent theoretical work by Audrin and Audrin (2023) advances beyond these limitations with their "digital emotional intelligence" (dEI) framework, integrating emotional intelligence and digital competence. Their empirical testing demonstrates a significant positive relationship between trait emotional intelligence and digital attitudes ($r=0.43$, $p<0.01$), providing the first empirical foundation for developing augmentative systems that complement human emotional capabilities. However, their framework lacks explicit computational dimensions needed for AI-augmented systems, representing a gap this project addresses.

Konrad et al. 's (2020) youth-specific research reveals that digital emotional expression incorporates multimodal elements (emojis, timing, media sharing) that create categorical rather than continuous emotional signaling-a finding that challenges traditional emotional intelligence frameworks built on continuous emotional models.

Computational Approaches to Digital Expression

Computational linguistics offers systematic analytical tools for digital expression that extend beyond human perception capabilities. The Linguistic Inquiry and Word Count framework (Pennebaker et al., 2015) provides validated approaches for extracting psychological meaning from text that have been empirically linked to mental health outcomes. However, its development with Western, primarily English-speaking populations creates significant limitations for application in Singapore's multicultural context with its unique linguistic patterns.

While computational approaches offer analytical power, most existing frameworks have three limitations for real-time emotional support: (1) they typically require extensive training data unavailable in specialized contexts like youth peer support, (2) they often lack integration with psychological frameworks, and (3) they frequently fail to account for individual baseline variations in expression patterns.

Dubé et al.'s (2024) findings on how psychological traits shape digital expression further complicate computational approaches. Their research demonstrates that emotional intelligence positively correlates with emoji frequency ($r=0.37$, $p<0.01$) while attachment avoidance negatively correlates with emoji use in close relationships ($r=-0.29$, $p<0.05$). These relationships suggest that standardized interpretation frameworks without personalization will systematically misinterpret expressions from individuals with different attachment styles—a limitation for existing computational approaches.

Beyond textual analysis, Li et al. (2023) discovered that late adolescents use digital music sharing as an emotional regulation tool, with 72% of participants reporting sharing music specifically to communicate emotions they struggled to express textually. This finding highlights a significant limitation in current computational approaches that focus primarily on text analysis while ignoring multimodal expression patterns particularly relevant to youth communication.

Human-AI Collaboration in Support Contexts

The field of human-AI collaboration provides crucial insights for integrating computational systems into human-centered emotional support. Davenport & Kirby's (2016) framework distinguishes between five types of automation/augmentation approaches, with "collaborative intelligence" positioned as most appropriate for emotionally complex domains. Their work provides design principles for AI systems that enhance rather than replace human capabilities—a central principle for this project's approach.

Research on high-stakes decision contexts demonstrates that human decision-makers benefit most from AI assistance when systems provide explanations and appropriate confidence levels (Lai & Tan, 2019). However, their research also reveals a critical limitation: as AI accuracy increases, human scrutiny decreases—a particularly concerning finding for emotional support contexts where algorithmic errors could have significant consequences.

Recent research by Kumar et al. (2025) on human creativity in the context of large language models reveals a concerning "assistance paradox": while AI assistance provided short-term performance boosts (17.3% improvement in ideation tasks), participants showed reduced independent performance (-8.7%) when working without assistance after extended AI use. This research poses fundamental questions about the long-term impacts of AI assistance on supporters' native abilities to interpret digital emotional expressions—questions this project's design must explicitly address.

The Gap: Integrating Approaches for Youth Support

The literature review reveals three gaps this project addresses:

First, while substantial work exists in individual domains, limited research integrates insights across emotional intelligence, computational linguistics, and human-AI collaboration specifically for youth support contexts. Existing computational approaches typically lack grounding in psychological frameworks, while psychological frameworks lack mechanisms for application to digital contexts.

Second, most existing AI systems in mental health focus either on fully automated support or clinical decision support for professionals, with limited attention to peer support contexts where non-professional supporters face unique challenges in digital emotional interpretation.

Third, the existing literature provides limited guidance on the optimal balance between AI assistance and human judgment in emotionally sensitive domains where both over-reliance and under-utilization of AI insights could negatively impact support quality.

This project addresses these gaps by developing an integrated approach that combines psychological frameworks, computational techniques, and human-AI collaboration principles specifically for youth peer support contexts in Singapore's multicultural environment.

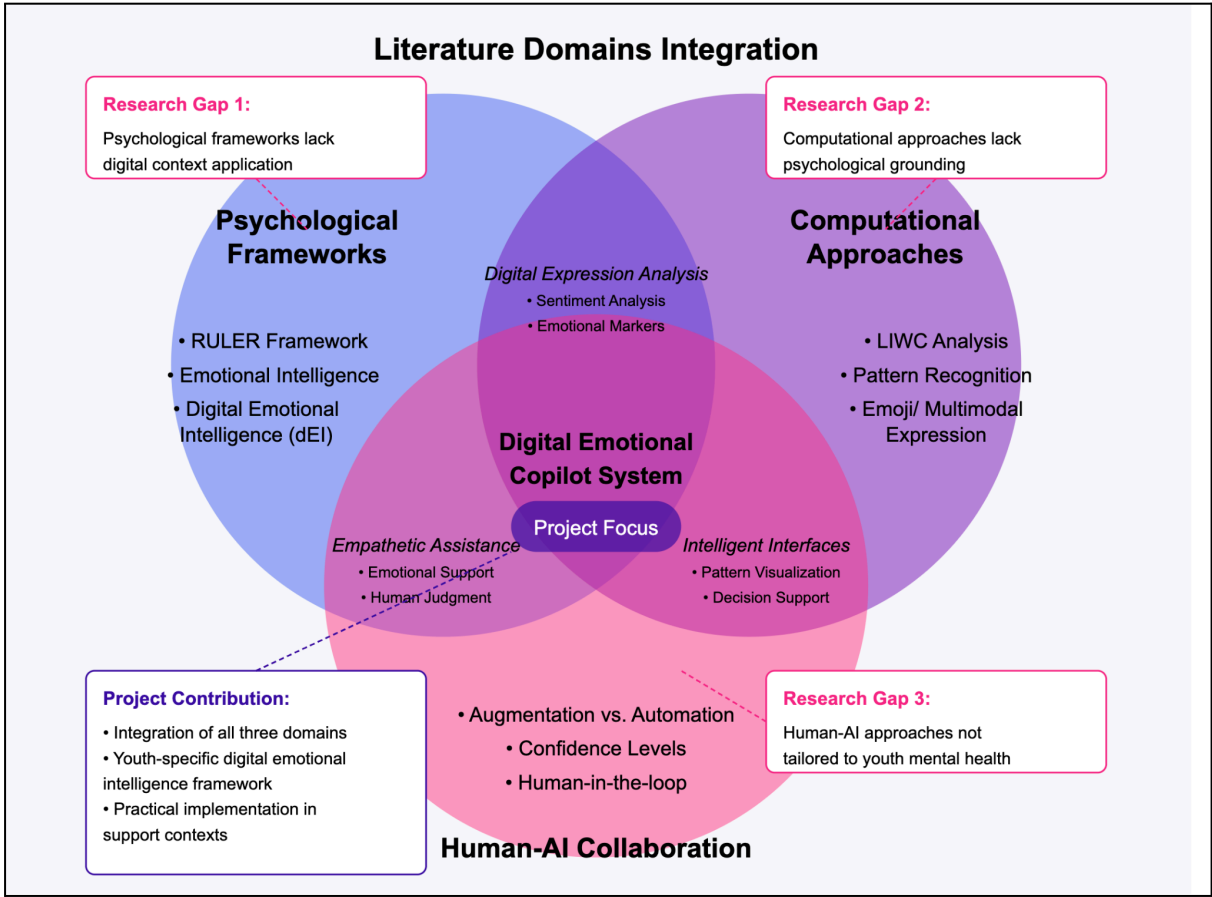


Figure 1: Integrated Literature Domains

Integrated literature domains guiding the Copilot system's theoretical foundation: emotional intelligence, computational linguistics, and human-AI collaboration.

Methodology

This project employed an iterative human-centered design methodology focused on developing a minimum viable product (MVP) for OTR's well-being mentors. The methodology combined qualitative user research, prototype development, and evaluation through multiple feedback cycles.

Research Design Overview

This project employed an iterative human-centered design methodology focused on developing a minimum viable product (MVP) for OTR's well-being mentors. This approach was selected over traditional waterfall development specifically because Singapore's youth mental health context required continual stakeholder feedback to ensure cultural relevance and practical utility. The methodology combined qualitative user research, prototype development, and evaluation through multiple feedback cycles aligned with the project's three guiding hypotheses.

The project followed four structured phases: Discovery, Definition & Design, Development, and Implementation. Each phase incorporated stakeholder feedback to balance academic rigor with the practical constraints of the capstone project timeline.



Figure 2: Human-Centered Design Process

The four-phase human-centered design process employed in the project, including Discovery, Definition & Design, Development, and Implementation.

Discovery Phase (January 15 - January 31, 2025)

The discovery phase focused on understanding the context, stakeholders, and specific challenges faced by well-being mentors at OTR. This phase utilized multiple complementary research methods:

Semi-structured interviews: We conducted in-depth interviews with 3 well-being mentors and 1 OTR leadership team member to understand their experiences, challenges, and workflows. Semi-structured interviews were chosen over structured interviews to allow for deeper exploration of emergent themes, while providing more methodological consistency than unstructured approaches. Interviews followed a protocol focusing on identifying pain points in digital emotional interpretation, current workarounds, and opportunities for technological assistance.

Literature review: A comprehensive review across psychological frameworks, computational approaches, and human-AI collaboration principles established the theoretical foundation for the system. The review specifically focused on digital emotional intelligence, computational approaches to text analysis, and effective human-AI collaboration models to inform design principles.

Definition & Design Phase (February 1 - February 19, 2025)

During the definition and design phase, I conducted two structured affinity mapping sessions with project stakeholders to organize research findings into thematic clusters. These sessions surfaced recurring challenges and opportunities, resulting in a prioritized list of user needs and system requirements. In addition, I developed detailed user journey maps to document the supporter experience during digital interactions. These maps helped identify pain points and moments where technological intervention would be most effective.

Drawing from these insights, I specified the system's functional and non-functional requirements, categorizing them as "must-have," "should-have," and "nice-to-have" based on their projected impact and feasibility. I then created low-fidelity wireframes for the copilot interface, refining them through multiple iterations informed by feedback from OTR supporters. These wireframes evolved into interactive prototypes that simulated core system functionality. To ensure ethical integrity, I also conducted a structured ethical impact assessment, identifying risks related to privacy, autonomy, and algorithmic bias, and proposing mitigation strategies accordingly.

This phase produced several key outcomes that laid the foundation for subsequent development. A detailed user journey map was created to highlight points for digital intervention across a typical support session. Based on this, a prioritized list of functional requirements for the minimum viable product (MVP) was established, focusing on the most impactful and feasible features. Low- and medium-fidelity prototypes of the copilot interface were also developed, providing a tangible basis for user feedback and iteration. An ethical framework was formalized to guide responsible system design and mitigate potential risks. Finally, integration specifications were drafted to ensure future compatibility with OTR's existing platform infrastructure.

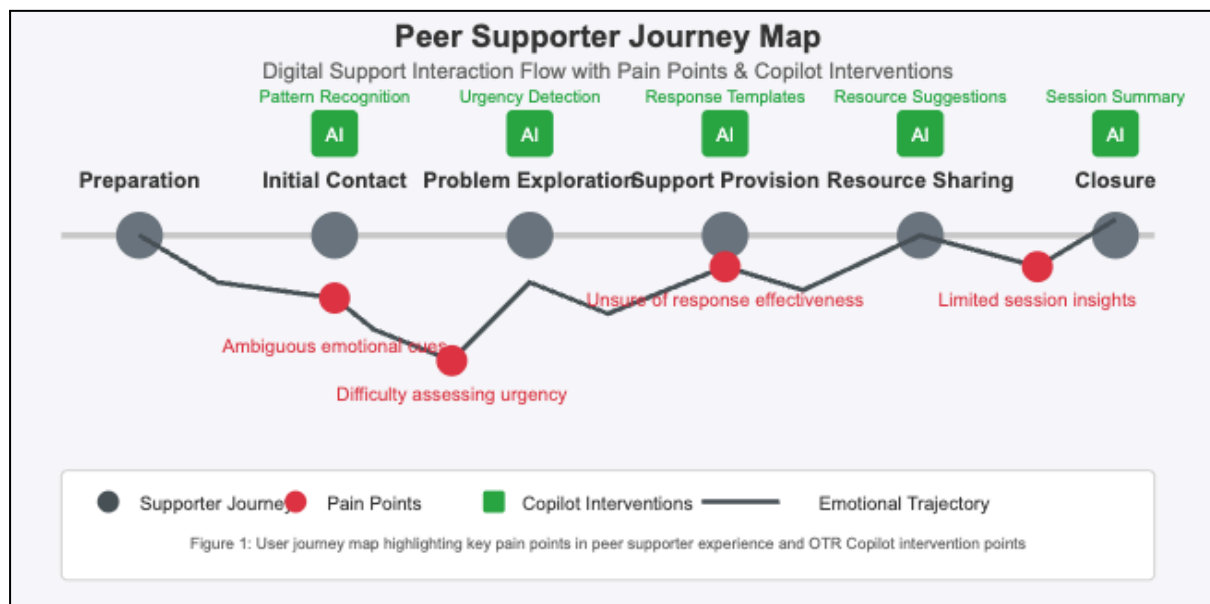


Figure 3: User Journey Map

User journey map highlighting key pain points and opportunities for digital intervention across a typical support session.

Development Phase (February 20 - April 15, 2025)

The development phase followed an iterative approach structured around three distinct releases, each shaped by ongoing stakeholder feedback. The first release, the Alpha version launched on March 14, 2025, focused on establishing core system functionality. This included developing the system architecture and backend, integrating chat system touchpoints, implementing user authentication and privacy safeguards, and constructing basic interface components. The Alpha version underwent internal validation through heuristic evaluation and privacy compliance checks in consultation with OTR's leadership team.

The second release, Beta v1, was completed on April 1, 2025. This version introduced expanded capabilities such as pattern recognition for emotional signals in text, message timing and frequency analysis, heuristic-based emoji interpretation, and preliminary urgency detection algorithms. A prototype interface for visualizing AI-generated insights was also introduced. Testing for Beta v1 involved three university students using a structured protocol with synthetic data designed to simulate real-world support conversations.

The final release, Beta v2, was delivered on April 15, 2025, incorporating refinements based on feedback from earlier testing. Key enhancements included improved algorithms, more intuitive visualizations of supporter insights, urgency detection augmented with confidence scoring, and cultural-linguistic adaptations specific to Singaporean youth. This version also featured linkage to OTR's internal knowledge base, allowing system-generated suggestions to be grounded in familiar content. Beta v2 was evaluated in realistic support scenarios involving mentors and simulated seekers, testing the system's responsiveness to a broad spectrum of emotional expressions.

Throughout the development phase, I maintained an agile, weekly refinement cycle that enabled continuous integration of user feedback and responsive iteration. The overarching design principle

emphasized augmentation over automation, ensuring that the system functioned as a supportive tool that enhanced-rather than supplanted-human empathy and interpretive judgment.

Implementation & Evaluation (April 15- April 17 2025)

The implementation and evaluation phase adopted a pragmatic approach suited to the constraints of an individual undergraduate capstone project. The aim was to gather useful feedback within a limited timeframe and resource base.

The technical evaluation involved manual testing of the core functionality across different web browsers to ensure cross-platform performance. Response times for key features were tracked during normal usage scenarios, and simple integration tests were run using sample API endpoints that mirrored those of OTR's system. For pattern recognition assessment, I tested the system using fifteen sample conversations that featured a range of emotional expressions. These were compared against outputs generated by the pre-trained model `mrm8488/t5-base-finetuned-emotion` from HuggingFace to identify major misclassifications or blind spots in the copilot's detection capabilities.

The user experience evaluation was structured around 60-minute sessions with three to five well-being mentors from OTR. Each session began with a brief system introduction lasting about ten minutes, followed by a guided walkthrough of two to three key features. Participants were then invited to independently explore the system using two pre-written support scenarios. The final segment of the session consisted of a feedback discussion aimed at surfacing usability insights and suggestions for improvement.

To collect feedback systematically, I administered a post-test questionnaire featuring five-point Likert scale ratings on perceived usefulness and ease of use. In addition to the quantitative data, I posed open-ended questions to elicit qualitative feedback on potential workflow improvements. Observational notes were taken throughout the sessions, and audio recordings of the discussions were made with participant consent to support subsequent analysis.

Prototype testing was conducted on April 17, with available OTR well-being mentors participating in the session. This test focused on core system functionality and gathered initial reactions to the interface, as well as suggestions for refinement. One key aim was to determine whether the system addressed the core pain points identified earlier in the research phase.

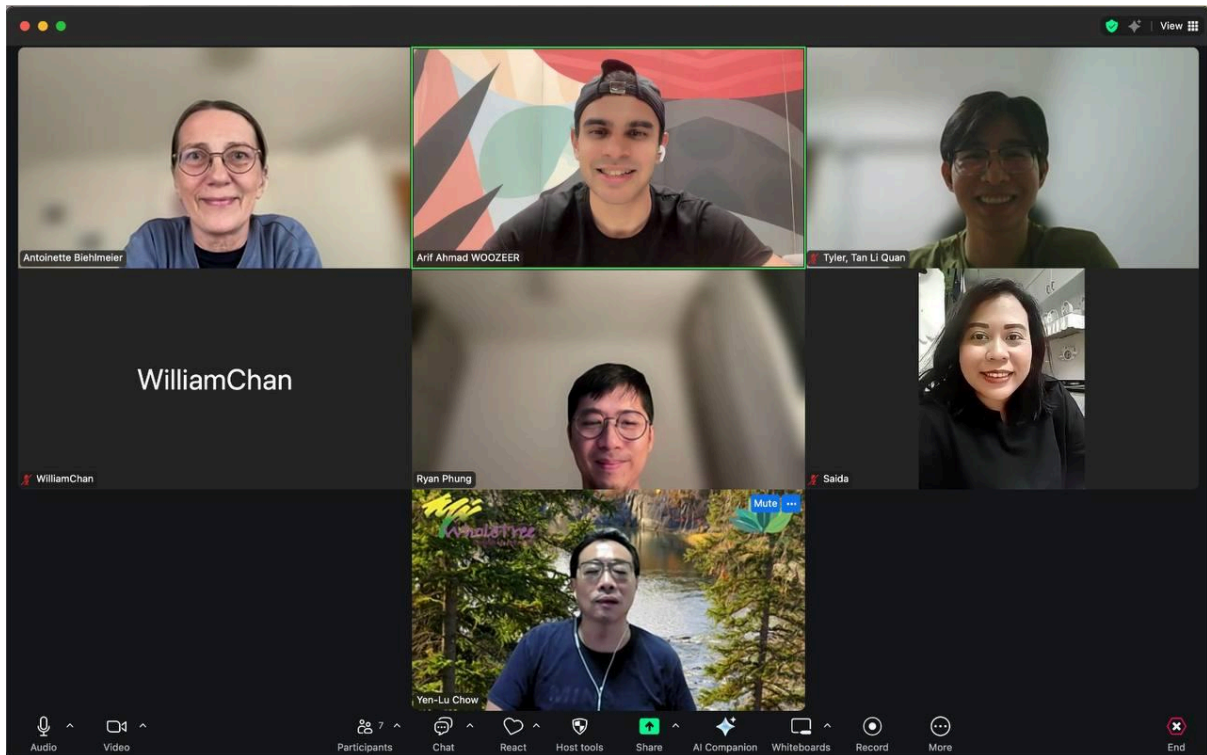


Figure 4: User Testing Session with OTR Mentors

Screenshot from the April 17, 2025 testing session conducted over Zoom, showing the project lead with OTR mentors and stakeholders. This collaborative session provided structured feedback on usability, AI assistance, and emotional interpretation features, and was key to validating the Copilot system's practical value.

Methodological Challenges and Adaptations

Several practical challenges emerged throughout the course of the project, each requiring specific adaptations to maintain the integrity and feasibility of the work. The original scope proved ambitious for a solo undergraduate project, prompting a focused prioritization strategy that concentrated efforts on developing core functionalities tied to the most pressing user needs identified during early-stage research.

Access to OTR well-being mentors was limited due to scheduling constraints, necessitating a flexible approach to interviews and testing. Sessions were carefully planned to maximize the value of participants' time, often combining multiple forms of feedback collection within a single engagement.

Technical constraints also shaped the development process. Certain advanced features related to pattern recognition were initially envisioned but proved too complex to implement within the available timeframe and resource limits. To address this, I leveraged established libraries and frameworks to ensure reliability while minimizing development overhead.

Privacy concerns presented another constraint, especially given the sensitive nature of support conversations. Without access to actual user data, I constructed representative synthetic conversations modeled on anonymized patterns described by OTR staff, thereby maintaining both ethical integrity and contextual relevance.

Full integration with OTR's technical infrastructure was not feasible within the scope of this project. As a result, I designed the copilot as a standalone system, ensuring it had clearly defined API specifications to enable future integration without requiring access to proprietary backend systems.

These adaptations reflect the realities of conducting an independent capstone project while maintaining fidelity to user-centered design principles and the overall objective of delivering a meaningful tool for OTR's well-being mentors.

System Design & Implementation

This section details the technical architecture, design decisions, and development process for the digital emotional expression copilot system. The design was informed by both theoretical foundations and practical user needs identified in the research phase.

System Architecture

The OTR Copilot system was designed with a modular architecture consisting of three primary components:

1. Analysis Engine

The analysis engine serves as the core component of the OTR Copilot system, responsible for processing conversation data and identifying emotional expression patterns. Built using JavaScript and Node.js, the engine includes several specialized modules. The text analysis module processes incoming textual content to detect emotional indicators, sentiment shifts, and potentially concerning language. Alongside this, a pattern recognition module monitors changes in communication frequency, response timing, and overall conversational flow to identify disruptions or escalations. An emoji and symbol interpreter analyzes nonverbal cues such as emoji usage and digital symbols as proxies for emotional states. To enhance contextual accuracy, a baseline tracker establishes and updates individualized communication norms for each user, enabling the system to detect deviations that may signal emotional changes.

2. Visualization Interface

The visualization interface is a Flutter-based front-end designed to deliver system insights to well-being mentors in an intuitive and non-disruptive manner. It includes a dashboard view that provides at-a-glance summaries for active support conversations, offering mentors a quick overview of emotional dynamics. A pattern timeline component allows mentors to visualize fluctuations in emotional expression over time, enhancing their situational awareness. Confidence indicators accompany system-detected patterns, giving users a sense of how reliable the AI-generated insights are. Additionally, the interface presents contextual suggestions-interpretative prompts that are meant to support, not override, the mentor's own judgment, thus preserving human empathy as the primary mode of support.

3. Integration Layer

The integration layer functions as a lightweight API service, enabling secure communication between the copilot system and OTR's existing platform. This component handles key backend tasks necessary for seamless interoperability. It manages user authentication and authorization protocols to ensure that access to sensitive data is properly controlled. Secure data transfer mechanisms are employed to protect user privacy and maintain confidentiality during interactions. Event synchronization functions ensure that relevant data flows in real time between systems, preserving continuity in user experience. Finally, privacy controls and consent management are built into this layer, allowing for explicit user control over data sharing and system usage, in line with ethical standards for digital mental health tools.

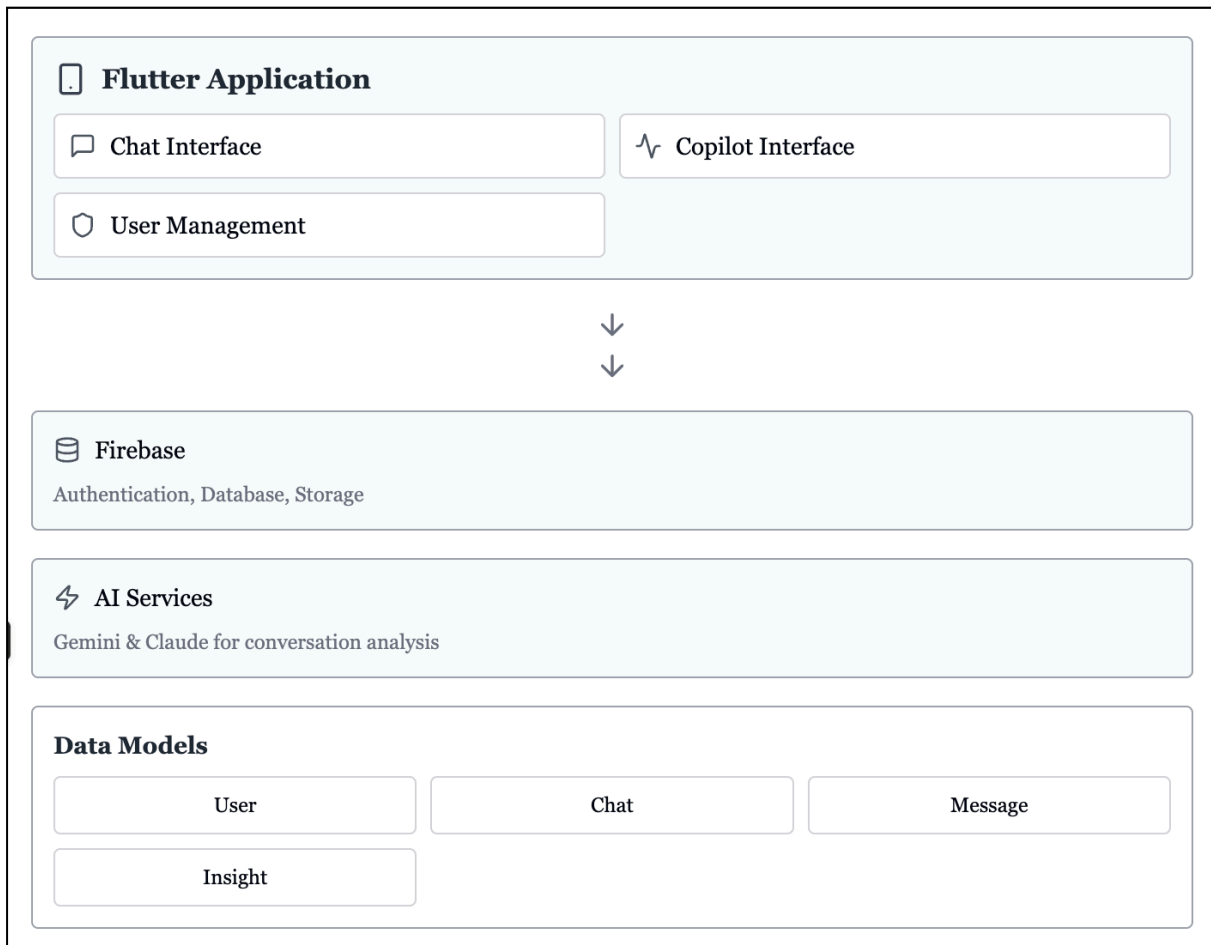


Figure 5: System Architecture Diagram

Modular architecture of the OTR Copilot system, comprising the analysis engine, visualization interface, and integration layer.

Database Schema

The system uses Firebase Firestore as its primary database, with the following collection structure:

- **Users Collection**
 - **user_id** (string): Unique identifier
 - **role** (string): Supporter or seeker
 - **preferences** (map): Coping strategies, preferred resources
- **Chats Collection**
 - **chat_id** (string): Unique conversation ID
 - **participants** (array): List of user IDs
 - **messages** (subcollection):
 - **sender_id** (string)
 - **timestamp** (datetime)

- **content** (string)
- **emojis_used** (array)
- **pattern_flags** (map): Emotional markers or urgency indicators
- **Resources Collection**
 - **resource_id** (string)
 - **title** (string)
 - **category** (string)
 - **url** (string)
 - **tags** (array)

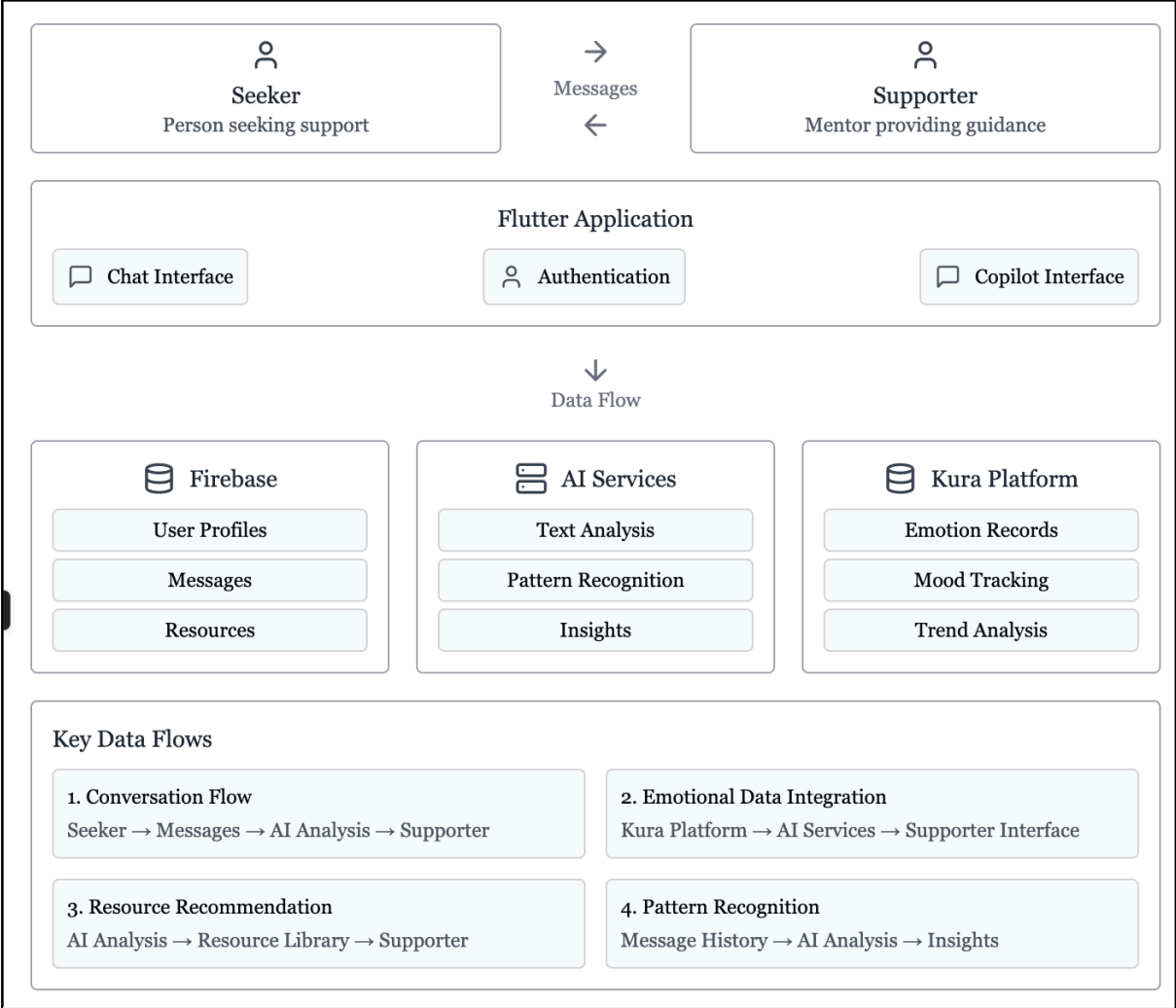


Figure 6: Data Flow Across Components

Illustration of the data flow between users, Firebase backend, AI analysis modules, and the front-end Copilot interface.

Core Features

The OTR Copilot system comprises four main features designed to support mentors in interpreting emotional expressions in digital support contexts.

1. Digital Expression Pattern Recognition

This feature enables the system to detect significant shifts in communication that may reflect underlying emotional changes. Timing analysis is used to monitor fluctuations in message frequency and response latency, identifying periods of unusually delayed or rapid communication. In parallel, the system conducts language pattern monitoring, which tracks changes in vocabulary usage and emotional tone over time. Emoji usage patterns are also analyzed, with the system evaluating both the frequency and type of emojis used as indicators of emotional intensity or mood shifts. Together, these elements form a multifaceted view of emotional expression that goes beyond the text alone.

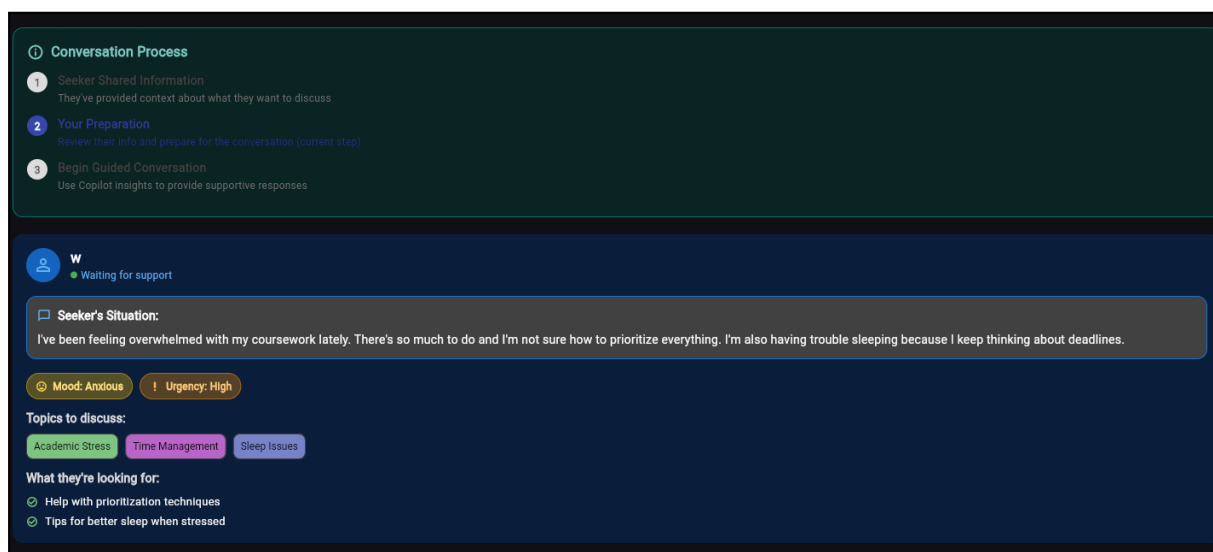


Figure 7: Pattern Recognition Dashboard

Copilot's pattern recognition interface showing emotion analysis and emotional intensity before commencing a synchronous session.

2. Context-Aware Analysis

To assist mentors in interpreting ambiguous messages, the system provides context-sensitive analysis grounded in conversation history. Messages are assessed not in isolation but in relation to past interactions, allowing the system to surface relevant historical patterns. This is further supported by a baseline comparison feature, which contrasts the current communication against each individual's established norms to flag unusual behavior. A topic tracking function identifies recurring themes or discussion topics, offering mentors deeper insights into the seeker's concerns and emotional patterns across multiple sessions.

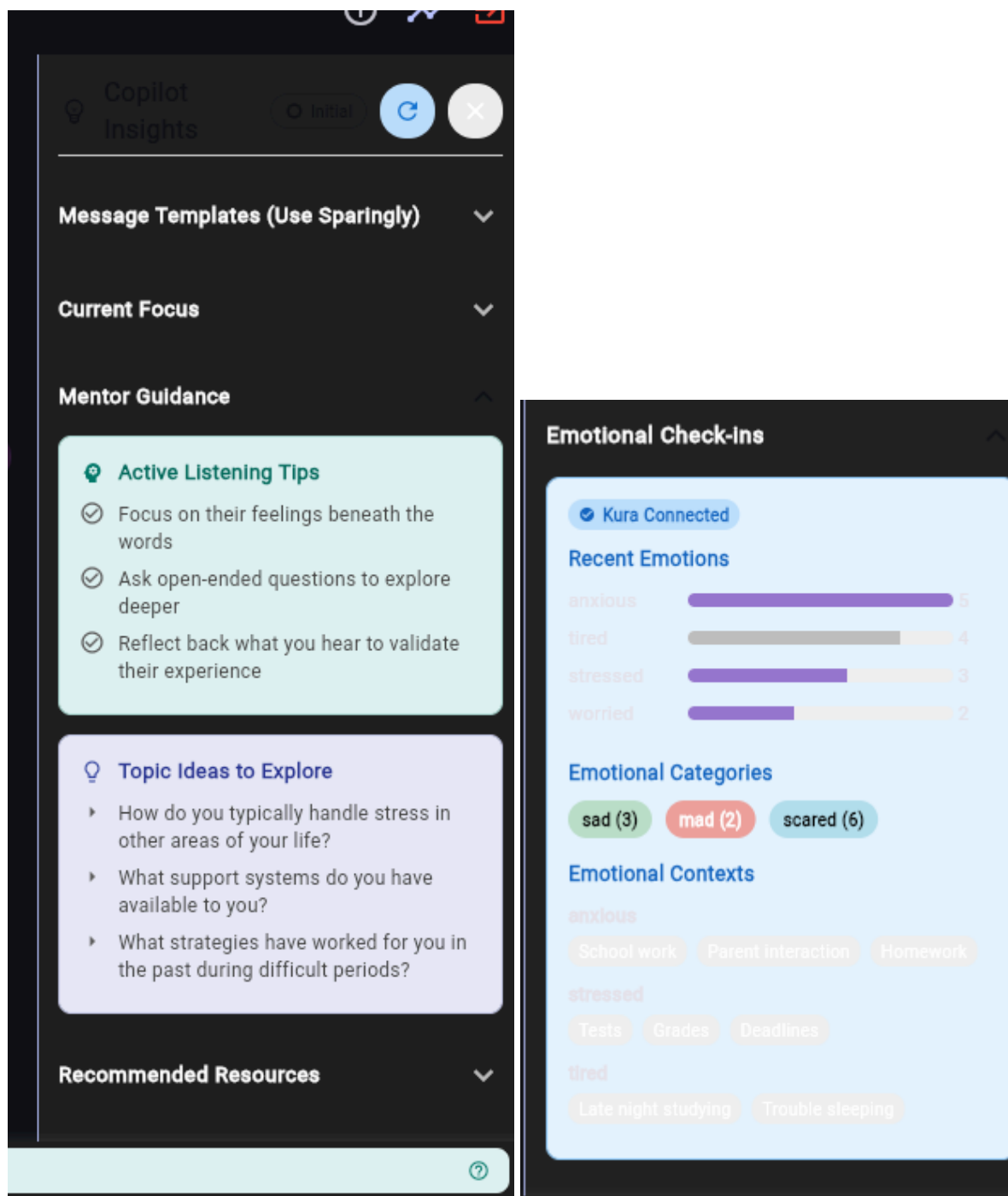


Figure 8: Context Analysis Panel

Context-aware analysis panel that visualizes baseline comparisons, recurring conversation topics, and emotional consistency over time.

3. Message Templates

This feature provides structured, customizable message suggestions aimed at improving the quality and empathy of mentor responses. The reflective approach includes templates that prompt mentors to help seekers articulate and reflect on their feelings, such as inviting them to describe how anxiety manifests physically. The analytical approach provides practical guidance based on evidence-based techniques, including suggestions related to stress management methods like structured breathing

exercises or the Pomodoro technique. The personalized approach adapts messages to align with the seeker's stated preferences or coping strategies, ensuring that support feels tailored and relevant. Each template is accompanied by an embedded reminder encouraging mentors to adapt the wording to suit their authentic voice, thus maintaining relational trust and conversational naturalness.

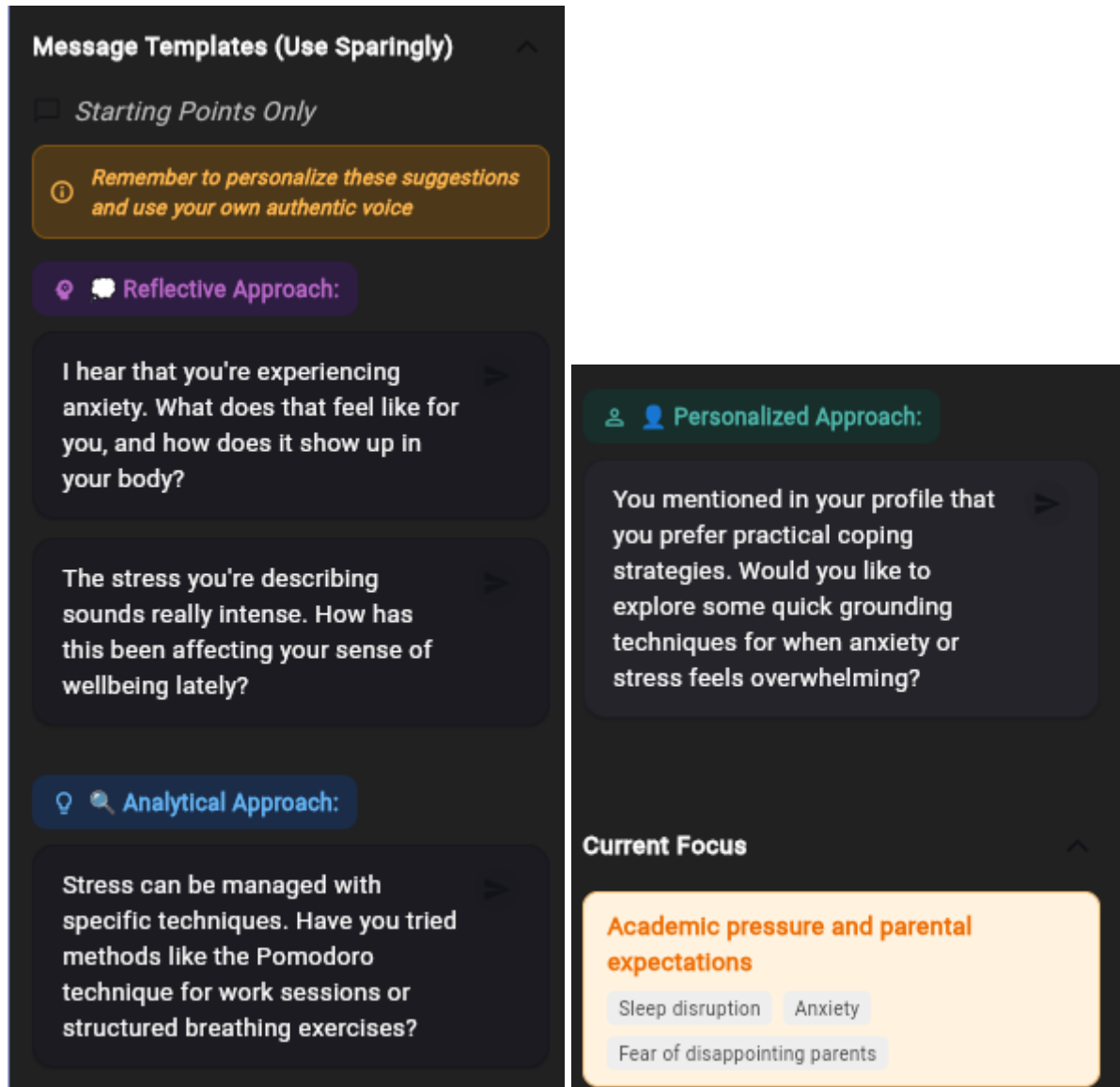


Figure 9a: Message Template Interface

Message suggestion interface providing customizable templates and embedded reminders for mentors to preserve their authentic voice.

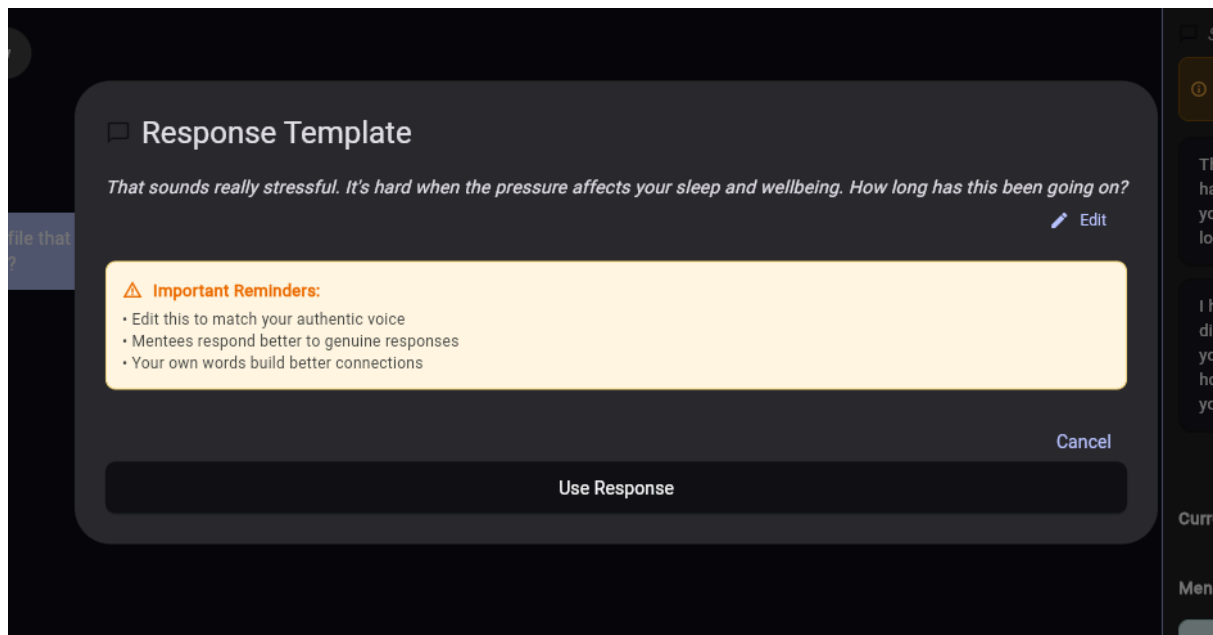


Figure 9b: Message Template Interface

Message suggestion interface includes pre-emptive nudges to ensure mentors use their authentic voice.

4. Resource Recommendations

To complement mentor support, the system also provides targeted resource suggestions based on ongoing conversation content. Through context-based matching, resources are recommended in real-time based on identified topics within the dialogue. These suggestions are further personalized according to the mentor's level of experience, ensuring that more advanced resources are surfaced for seasoned mentors while more foundational materials are presented to newer supporters. Quick-access integration allows mentors to engage with these resources without leaving the support interface, streamlining the overall user experience.

The interface showcases a suggested response alongside guidance encouraging mentors to personalize the message. This design emphasizes augmentation over automation, supporting relational nuance in emotionally sensitive contexts.

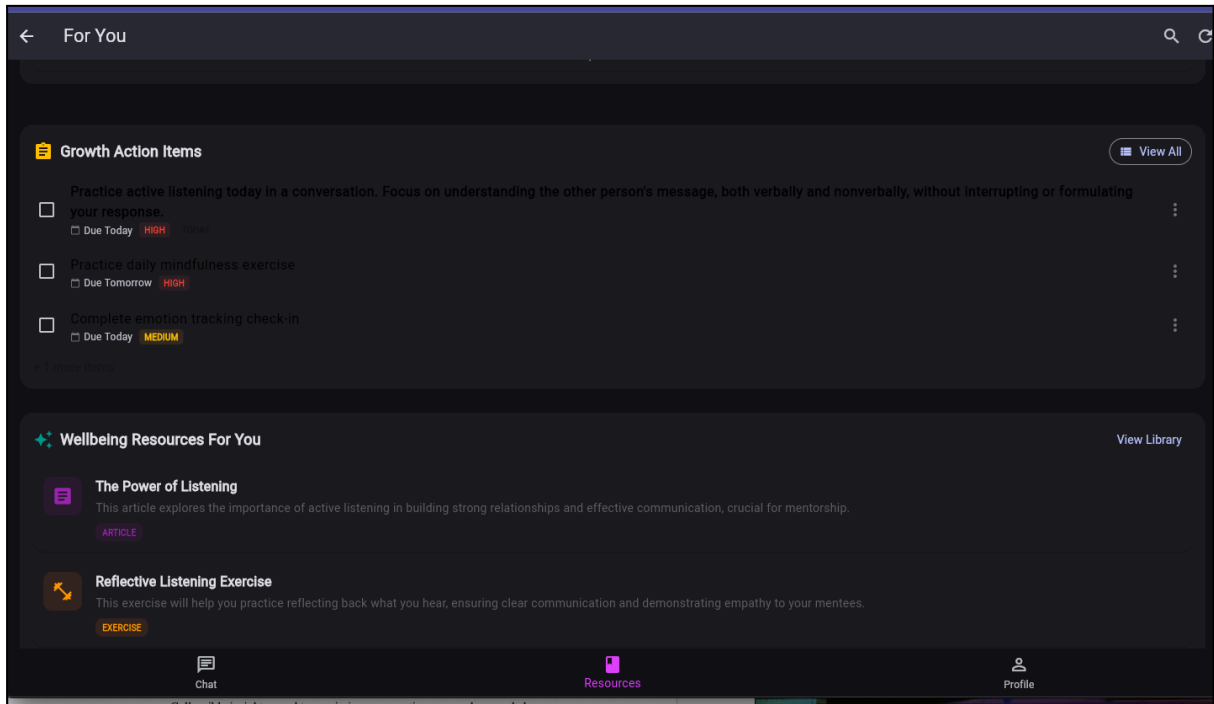


Figure 10: Resource Recommendation Panel

Supporter-facing resource panel recommending contextually relevant self-care or psychoeducational content based on the current conversation.

User Interfaces

The interface design of the OTR Copilot system followed minimalist principles, ensuring that well-being mentors could access key insights without feeling overwhelmed. The goal was to offer meaningful support without distracting from the primary task of engaging empathetically with seekers.

Supporter Interface Components

The main interface centers on a split-screen layout, featuring the real-time conversation on the left and the Copilot insights panel on the right. This layout allows mentors to monitor AI-generated cues alongside ongoing dialogue. To accommodate different working styles, the insights panel is collapsible, enabling users to maximize conversation space when needed. Visual indicators subtly signal the availability of new insights, ensuring that mentors remain informed without experiencing disruptive notifications.

A centralized dashboard provides an overview of active support conversations. It includes emotional status indicators at a glance, giving mentors a high-level understanding of each session's emotional trajectory. The dashboard also allows for quick access to recent conversations and relevant support resources.

Following each session, a summary interface presents visualizations of emotional expression patterns throughout the conversation. This includes topic tracking displays that highlight the evolution of key themes, as well as an emotional expression summary to support post-session reflection and documentation.

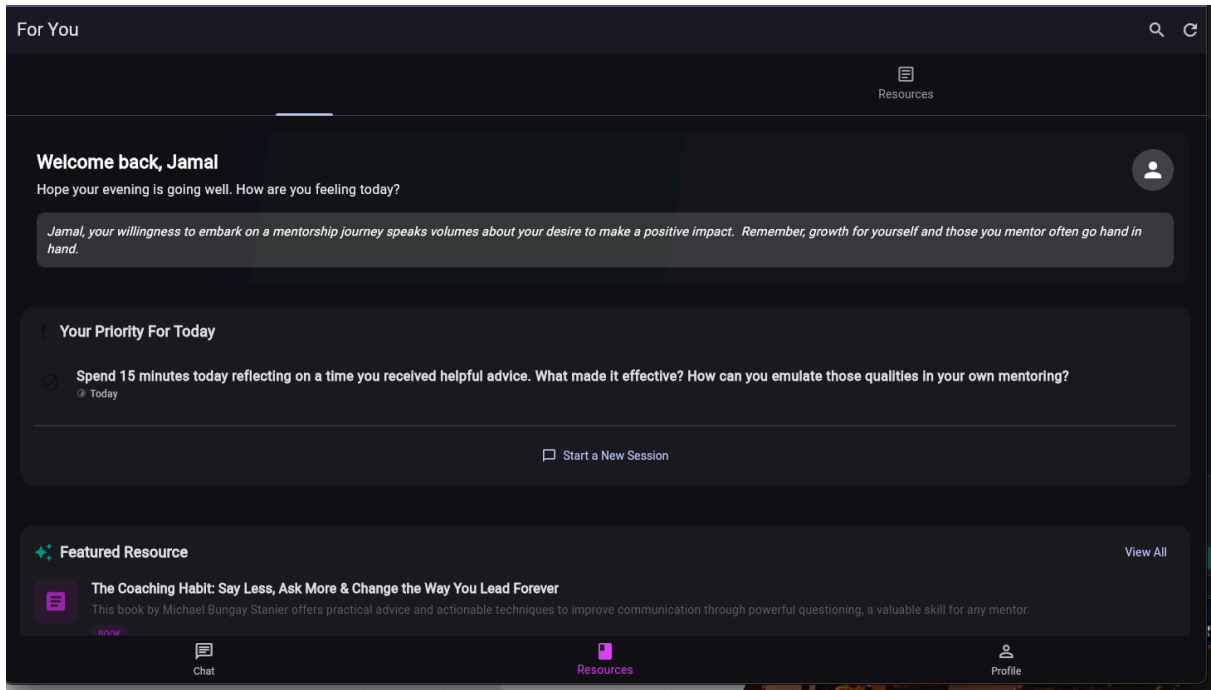


Figure 11: Supporter Chat Interface

Main supporter interface displaying a real-time conversation alongside Copilot-generated insights and emotional cues.

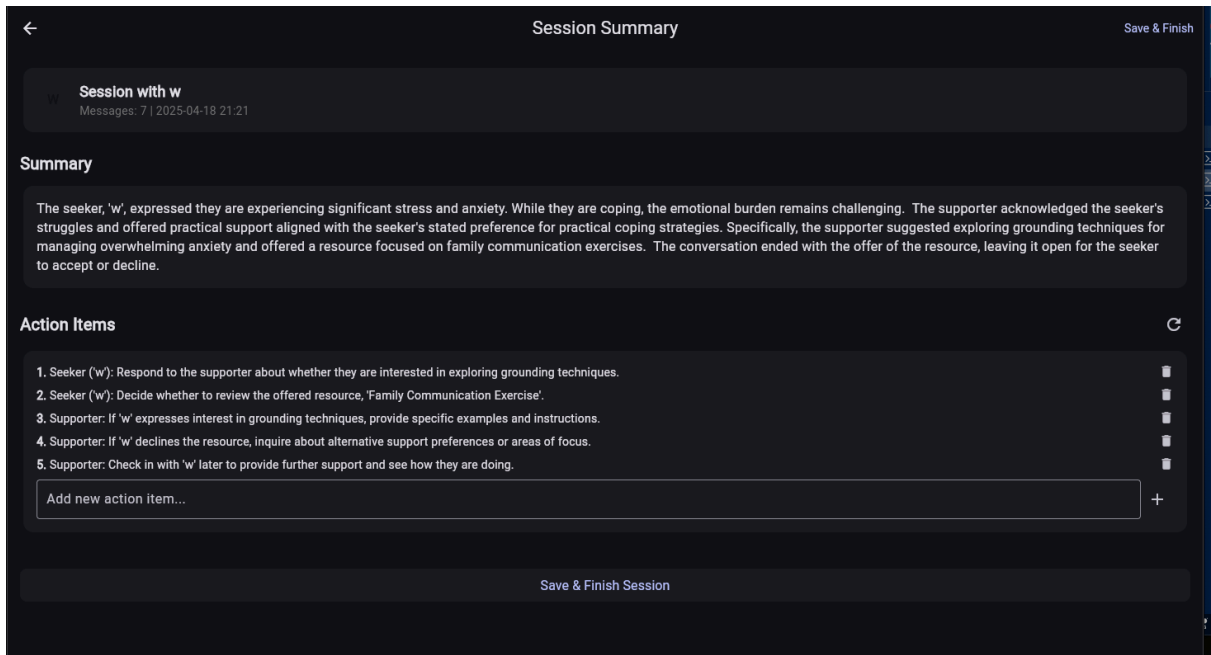


Figure 12: Post Session Interface

Main supporter interface displaying summary and AI-generated action items,

Development Process

The development of the OTR Copilot system unfolded over three iterative phases, each building on the progress and feedback of the previous stage.

The first phase, **Prototype Development**, ran from February 20 to March 14. During this period, I established the foundational system architecture and mapped out the core data flow between components. Initial efforts focused on implementing basic text analysis capabilities and developing a minimal viable dashboard interface to support early testing. To simulate integration with OTR's infrastructure, mock endpoints were created, enabling the system to function independently while maintaining structural compatibility with future integrations.

The second phase, **Core Functionality Development**, took place from March 15 to April 1. During this time, enhancements were made to the pattern recognition algorithms, refining the system's ability to detect emotional signals. The urgency detection module was also upgraded to improve sensitivity and reliability. Additionally, I developed interactive visualizations to help mentors interpret system insights, and implemented feedback collection mechanisms directly within the interface to support ongoing refinement.

The third phase, **Refinement and Enhancement**, spanned April 2 to April 15. In this final sprint, I incorporated feedback from earlier testing to improve overall usability and interface intuitiveness. The accuracy of pattern recognition algorithms was further refined, and the performance of the system was optimized to ensure a smooth user experience. These refinements were crucial in preparing the prototype for the implementation and evaluation phase.

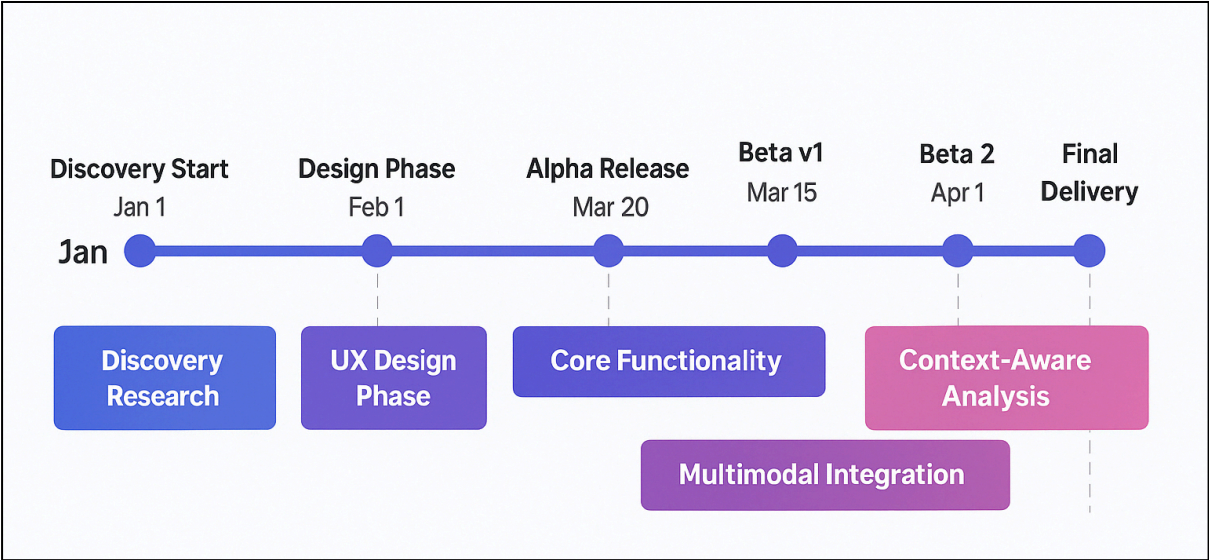


Figure 13: Development Timeline

Development timeline illustrating key phases and milestones across the 3-month implementation period using an iterative design approach.

Technical Implementation Details

The OTR Copilot system was developed using a technology stack chosen for its balance of feasibility, scalability, and developer familiarity. This section provides detailed information about the technical approaches and design decisions that informed the implementation of the pattern analysis components, with particular attention to Singapore-specific linguistic considerations.

Core Technology Stack

For front-end development, I employed Flutter, which enabled efficient cross-platform application design and rapid interface prototyping. The backend infrastructure was supported by Firebase, which provided services for user authentication, real-time database storage via Firestore, and serverless function execution. These tools allowed for a streamlined development process without the overhead of managing a custom server, while maintaining the flexibility needed to incorporate specialized linguistic analysis components.

The system also leverages Google's Gemini 2.5 Flash model for aspects of the message template generation and context-aware suggestions.. This integration of a large language model (LLM) allows for sophisticated natural language generation capabilities within resource constraints.

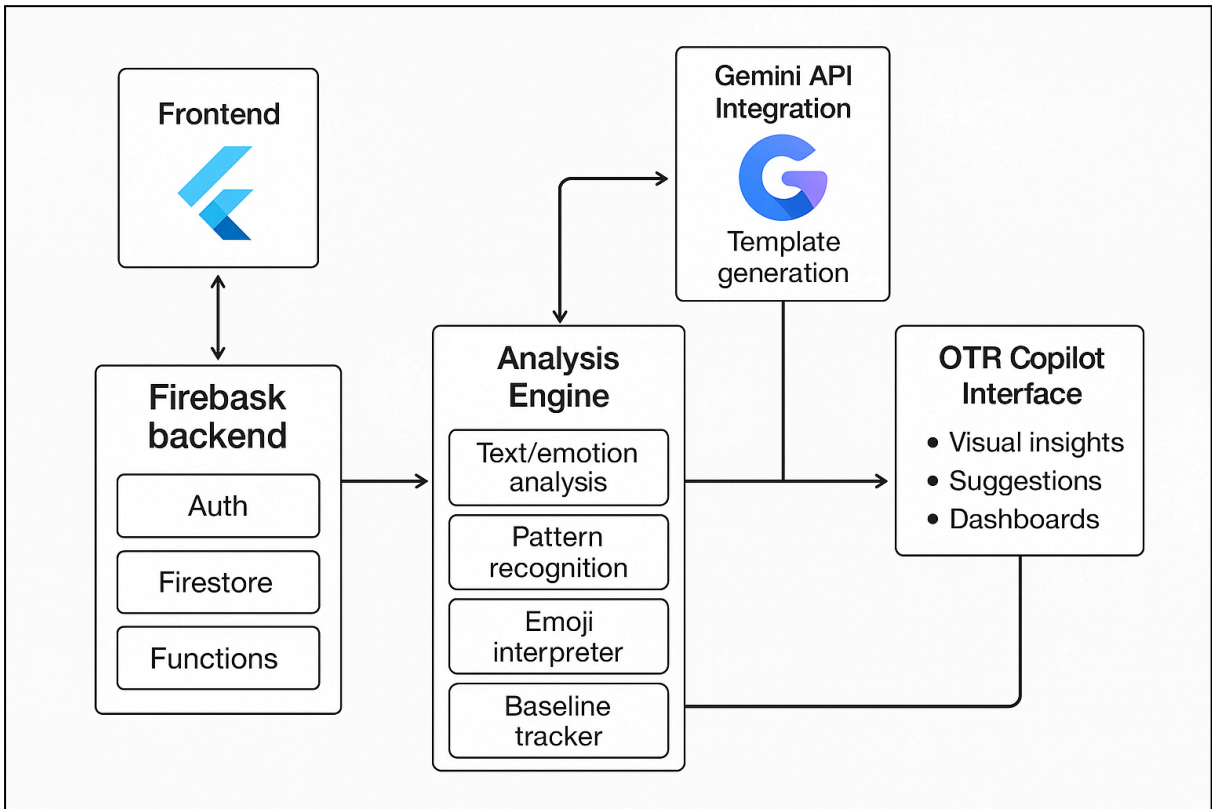


Figure 14: System Architecture Overview

System architecture of the OTR Copilot, showing the interaction between frontend, backend, analysis engine, and AI template generation.

Analysis Engine Implementation

1. Text Analysis Module

The text analysis module employs several computational approaches to detect emotional indicators in text, with specific adaptations for Singapore English contexts:

Lexicon-Based Sentiment Analysis with Cultural Adaptation: For sentiment detection, I implemented a lexicon-based approach similar to VADER (Valence Aware Dictionary and sEntiment Reasoner). This approach was selected because of its interpretability and ability to handle informal text containing emojis and slang-characteristics common in youth digital communication. The implementation required significant adaptation to account for Colloquial Singapore English (CSE) patterns, drawing insights from the Corpus of Singapore English Messages (CoSEM) (Gonzales et al., 2023).

The adaptation process involved examining patterns of emotional expression unique to Singapore English, such as the use of sentence-final particles like "sia" (Hafiz et al., 2024) and adverbial markers like "confirm" (Lim et al., 2024) that carry significant emotional weight in local contexts.

The following code snippet illustrates the implementation of these Singapore-specific adaptations in the text analysis module.

```

// Singlish-aware emotional text analyzer with cultural adaptation
class SingaporeEnglishAnalyzer {
  constructor() {
    // Base lexicon initialization
    this.baseEmotionLexicon = require('./lexicon/nrc_emotion.json');

    // Singapore-specific modifiers with emotional intensity factors
    this.singlishModifiers = {
      // Sentence-final particles with emotional weighting
      "lah": { intensity: 1.2, confidence: 0.8, emotions: ['emphasis', 'assertion'] },
      "lor": { intensity: 0.8, confidence: 0.7, emotions: ['resignation', 'acceptance'] },
      "leh": { intensity: 1.1, confidence: 0.8, emotions: ['uncertainty', 'question'] },
      "sia": { intensity: 1.4, confidence: 0.9, emotions: ['surprise', 'emphasis'] },
      "hor": { intensity: 1.0, confidence: 0.7, emotions: ['seeking_agreement'] },

      // Intensifiers with emotional implications
      "very": { intensity: 1.3, confidence: 0.9, emotions: ['emphasis'] },
      "damn": { intensity: 1.5, confidence: 0.9, emotions: ['emphasis', 'frustration'] },
      "super": { intensity: 1.4, confidence: 0.8, emotions: ['emphasis', 'appreciation'] },
      "confirm": { intensity: 1.3, confidence: 0.8, emotions: ['certainty', 'emphasis'] },

      // Common code-mixed expressions with emotional connotations
      "bojio": { intensity: 1.2, confidence: 0.7, emotions: ['exclusion', 'disappointment'] },
      "sian": { intensity: 1.3, confidence: 0.9, emotions: ['boredom', 'frustration'] },
      "kiasu": { intensity: 1.2, confidence: 0.9, emotions: ['anxiety', 'fear_of_missing_out'] },
      "rabak": { intensity: 1.4, confidence: 0.7, emotions: ['chaotic', 'overwhelmed'] }
    },
  };

  // Contextual pattern detection for multi-word expressions
  this.contextualPatterns = [
    {
      pattern: /cannot\s+(already|liddat|like\s+that)/i,
      emotions: ['frustration', 'helplessness'],
      intensity: 1.4,
      confidence: 0.8
    },
    {
      pattern: /(confirm|sure)\s+(plus|plus\s+chop)/i,
      emotions: ['certainty', 'emphasis'],
      intensity: 1.5,
      confidence: 0.9
    },
    // Reduplication patterns common in Singapore English
    {
      pattern: /(\w+)[-s]\1/i,
      emotions: ['emphasis', 'intensity'],
      intensity: 1.3,
      confidence: 0.7,
      handler: this.handleReduplication
    }
  ];

  handleReduplication(match, text) {
    // Special handling for reduplication patterns (e.g., "long long time")
    // which intensify the emotional content of the repeated word
    const word = match[1].toLowerCase();
    const baseEmotion = this.getBaseEmotionForWord(word);
    return {
      emotions: baseEmotion ? baseEmotion.emotions : ['emphasis'],
      intensity: baseEmotion ? baseEmotion.intensity * 1.3 : 1.3,
      confidence: baseEmotion ? Math.min(baseEmotion.confidence * 1.1, 0.95) : 0.7
    };
  }

  analyzeText(text) {
    const tokens = this.tokenize(text);
    let emotionalProfile = this.computeBaseEmotions(tokens);

    // Apply Singapore-specific modifiers
    emotionalProfile = this.applySinglishModifiers(text, tokens, emotionalProfile);

    // Apply contextual pattern analysis
    emotionalProfile = this.applyContextualPatterns(text, emotionalProfile);

    // Calculate confidence scores
    const confidenceScore = this.calculateConfidence(emotionalProfile);

    return {
      dominantEmotion: this.getDominantEmotion(emotionalProfile),
      emotionalProfile: emotionalProfile,
      confidence: confidenceScore,
      culturalContextNotes: this.getCulturalContext(text, emotionalProfile)
    };
  }
}

```

Listing 1: Singapore English Adaptations - Code Snippet

This implementation demonstrates how cultural-linguistic patterns are systematically integrated into the emotional analysis pipeline, allowing for more accurate interpretation of expressions unique to Singapore's multicultural context.

These adaptations were important for accurate sentiment analysis, as standard English sentiment lexicons often miss or misinterpret these culturally specific emotional markers.

For example, expressions like "cannot already" might register as neutral or mildly negative in standard sentiment analysis but carry stronger emotional valence in Singaporean youth contexts. Similarly, the implementation recognizes intensifiers common in CSE such as "super" and reduplication patterns like "long long time" that modify emotional intensity in ways that differ from Standard English usage patterns.



Figure 15: Analysis Engine Component Map

Modular components of the Copilot's analysis engine, annotated with Singapore-specific adaptations such as code-switching awareness, emoji reduplication, and culturally unique emotional markers.

Emotion Classification with Cultural Context: To move beyond simple positive/negative sentiment and capture specific emotions, I integrated a categorical emotion detection system based on the wheel of emotions model. This implementation identifies eight primary emotions (anger, fear, anticipation, trust, surprise, sadness, joy, and disgust) with specific adaptations for cultural expression patterns found in Singapore youth communication:

1. Word-emotion association detection calibrated to CSE expressions
2. Contextual analysis of surrounding text with awareness of code-switching patterns

3. Intensity scoring based on linguistic intensifiers common in Singaporean youth communications

The emotion classification system was particularly attentive to the ways Singapore youth blend expressions from multiple languages, such as Hokkien-derived expressions like "(bo)jio" studied by Gonzales et al. (2023), which carries emotional connotations of inclusion/exclusion that might be missed by standard emotion detection systems.

Topic Detection for Youth Support Contexts: To identify recurring themes in conversations, I implemented a keyword-based topic detection system tailored to youth well-being concerns in Singapore. This approach was chosen over more complex topic modeling techniques like LDA due to the limited availability of training data in this context. The system:

1. Extracts key terms using TF-IDF weighting
2. Applies contextual understanding of Singapore education system terminology and youth concerns
3. Maps clusters to predefined topics relevant to youth well-being in Singapore's cultural context

For topic detection, I developed topic clusters specifically relevant to Singapore youth contexts, including academic stress (with terms associated with the local examination systems), family pressure terminology, and social belonging concerns expressed in culturally specific ways. This contextual adaptation was essential for making the system relevant to the actual concerns expressed by youth seeking support through OTR's platform.

2. Pattern Recognition Module

The pattern recognition module analyzes communication frequency, response timing, and conversational flow through several computational approaches adjusted for local communication patterns:

Temporal Pattern Analysis with Contextual Awareness: I implemented a statistical analysis approach to detect significant changes in message frequency and response timing. The algorithm processes conversational data to:

1. Calculate moving averages for response times within conversations
2. Determine standard deviations from baseline timing patterns for individual users
3. Identify acceleration or deceleration of message frequency
4. Detect timing patterns associated with emotional distress signals in Singapore youth communication

Research by Leimgruber et al. (2020) on discourse particles in CSE informed the implementation of timing pattern analysis. Their work demonstrates how timing and particle use interact in CSE digital communication, providing valuable insights for interpreting message timing in the emotional support context. The system takes into account the tendency in Singapore digital communication for rapid-fire exchanges during emotional episodes, a pattern identified during testing with OTR mentors.

Conversational State Tracking with Cultural Adaptation: To capture conversational flow patterns, I implemented a state-based model that tracks transitions between emotional states within conversations. This implementation was specifically designed to accommodate Singapore youth communication patterns by:

1. Mapping detected emotions from the Text Analysis Module to defined emotional states
2. Tracking transitions between emotional states with sensitivity to cultural expression patterns
3. Flagging unusual transitions or persistent emotional states based on established baselines
4. Recognizing culturally-specific emotional expression sequences common in Singapore youth digital communication

This approach allows the system to identify when conversations follow atypical emotional trajectories, potentially indicating escalation or concerning patterns. The state tracking system was calibrated to recognize emotional progression patterns observed in OTR's support conversations, including adaptation for Singapore-specific indirect expression of emotional distress that might be missed by systems designed for more direct expression styles.

3. Emoji and Symbol Interpreter

The emoji analysis component was implemented using a contextually-aware approach that recognizes Singapore-specific digital communication patterns:

Culturally-Informed Emoji Analysis: The system assigns sentiment and emotional values to emojis based on their usage patterns in Singapore youth communication. This implementation recognizes that emoji usage varies significantly across cultures and age groups, with Singapore youth demonstrating distinct patterns including:

1. Higher frequency usage of certain emotional emojis compared to international norms
2. Culturally-specific emoji combinations that carry localized meanings
3. Context-dependent shifting of emoji meanings based on surrounding text and patterns

Contextual Emoji Interpretation: I developed a rule-based system that adjusts emoji interpretations based on:

1. Context within the conversation and surrounding linguistic content
2. Common combinations and sequences of emojis in Singapore youth communication
3. Relationships between text and emoji usage, particularly in emotionally charged exchanges

The implementation recognizes Singapore-specific emoji patterns such as the use of multiple emojis for emphasis rather than changing meaning, and the tendency to use seemingly contradictory emojis to soften emotionally charged messages-patterns observed during interviews with OTR mentors and in sample conversations.

Usage Pattern Monitoring: The system tracks changes in emoji usage over time for individual users, implementing:

1. Baseline establishment for individual users' typical emoji frequency and type preferences
2. Detection of significant changes in emoji type and frequency that may signal emotional shifts

3. Classification of emoji functions (emotional expression, conversation management, tone modification) based on contextual usage

This component draws from research on digital emotional expression by Novak et al. (2015) and Kaye et al. (2017) that demonstrates how changes in emoji usage patterns can signal emotional state changes. The implementation includes adaptations for Singapore-specific digital expression norms documented in Gonzales et al.'s (2023) Corpus of Singapore English Messages (CoSEM) and the cultural emoji usage patterns identified by Leimgruber et al. (2020).

4. Baseline Tracker Module

The baseline tracker establishes and updates individualized communication norms for each support seeker:

Adaptive Baseline Calculation: I implemented an adaptive algorithm to establish baseline communication patterns for individual users. The implementation:

1. Weights recent conversations more heavily than older ones to track evolving emotional states
2. Requires a minimum number of interactions before establishing full baselines
3. Stores user-specific baseline metrics securely in the Firestore database
4. Adapts to individual communication style preferences that may reflect cultural or personal variation

Pattern Deviation Detection: Using the established baselines, the system:

1. Calculates statistical measures of deviation from typical patterns for each user
2. Applies different thresholds for different communication metrics based on their predictive value
3. Aggregates unusual patterns across multiple metrics to reduce false positives and improve detection accuracy

This approach enables personalized interpretation of emotional signals while maintaining privacy and security of user data. The baseline tracker is particularly important in Singapore's multicultural context, where communication norms vary significantly across different ethnic and cultural backgrounds—a diversity explicitly recognized in the CoSEM corpus metadata structure (Gonzales et al., 2023).

5. Prompt Engineering for Message Templates

An innovative aspect of the OTR Copilot system is the implementation of context-aware message templates using the Gemini 2.5 Flash model. This component required careful prompt engineering to generate appropriate, empathetic responses that maintain the natural voice of mentors while providing structured support.

Prompt Architecture for Context-Aware Templates: I developed a sophisticated prompt architecture that translates the emotional and contextual insights from the analysis engine into useful message suggestions:

Structured Context Framing: Each prompt to the Gemini model includes structured components that provide essential contextual information:

```
conversation_history: [last 5 messages with emotional tags]

current_emotional_state: [primary and secondary emotions detected]

conversation_topic: [identified topic clusters]

seeker_communication_style: [baseline metrics for the current
seeker]

urgency_level: [detected urgency indicators]
```

Culture-Specific Instruction Layer: The prompts incorporate specific instructions about Singapore youth communication norms, including:

- Recognize indirect expressions of distress common in Singapore youth contexts
- Consider cultural perspectives on academic pressure within Singapore's education system
- Maintain awareness of family dynamics common in Asian contexts
- Avoid Western-centric assumptions about youth independence and agency

Template Generation Parameters: Each prompt includes specific parameters that guide the generation of relevant, situationally-appropriate templates:

```
approach: ["reflective", "analytical", "personalized"]

tone: ["encouraging", "gentle", "matter-of-fact", "affirming"]

length: [ranges from brief to comprehensive based on context]

structure: [specific message structure based on OTR protocols]
```

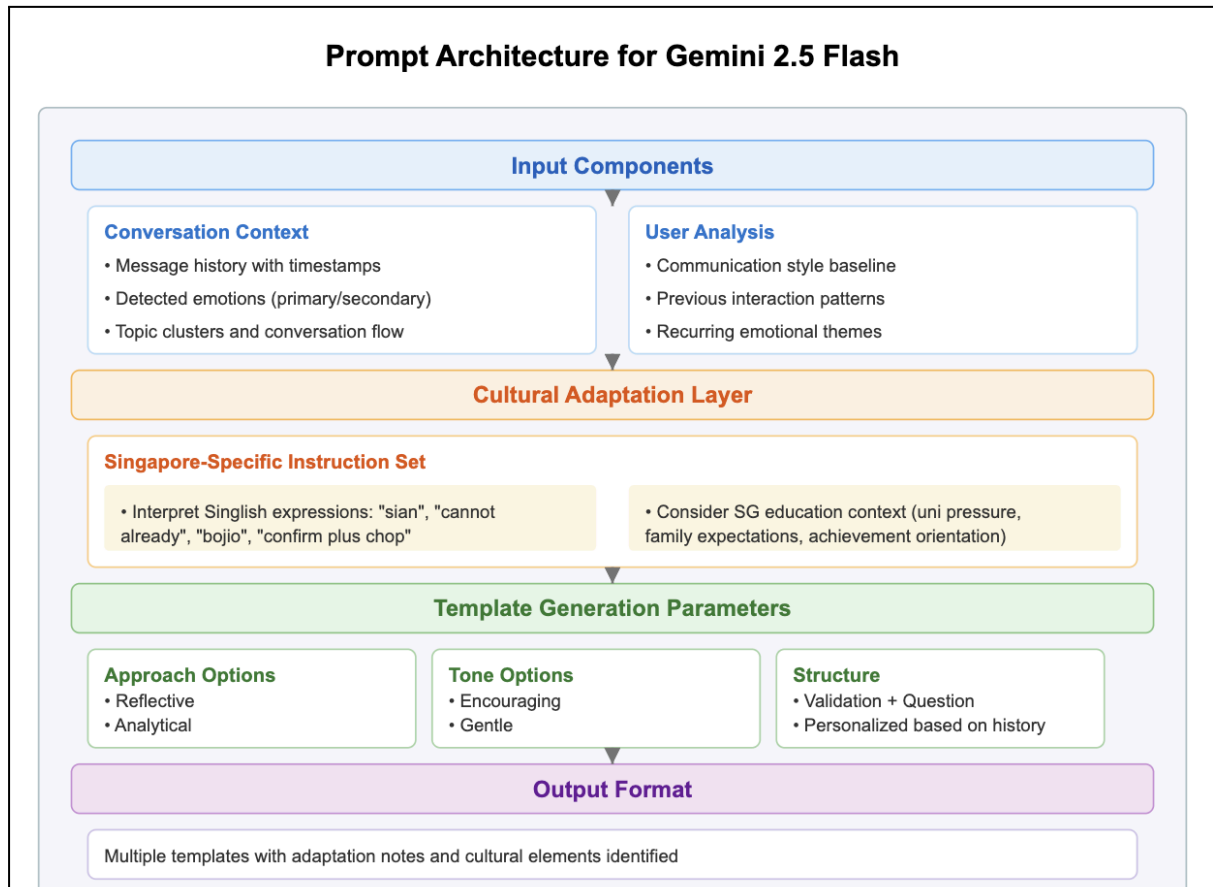


Figure 16: Prompt Architecture Schematic

Structure of dynamic prompt generation for Gemini 1.5 Pro, including cultural adaptation layers and response formatting.

Dynamic Prompt Construction: Rather than using static prompts, I implemented a dynamic prompt construction system that assembles prompt components based on the specific conversation context. This approach:

1. Analyzes the current conversation state from multiple dimensions
2. Selects relevant cultural, emotional, and situational context components
3. Constructs a tailored prompt that guides Gemini to generate appropriate templates
4. Includes specific examples of effective responses based on OTR's support practices

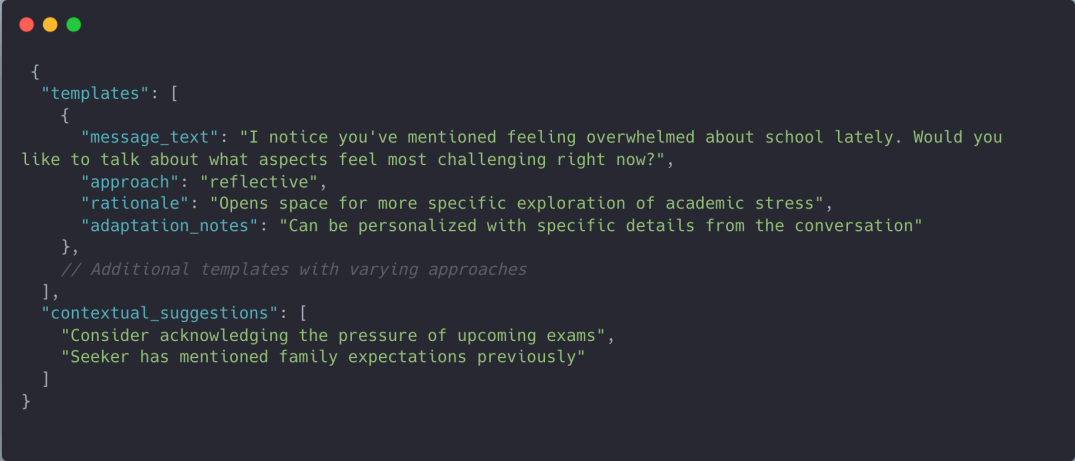
Parameter Tuning and Constraint Implementation: The prompt engineering process involved extensive experimentation with Gemini's parameters to achieve the right balance of creativity, relevance, and empathy:

1. **Temperature Optimization:** I tested various temperature settings to balance creativity with reliability, ultimately implementing a dynamic temperature adjustment mechanism that varies based on conversation urgency (lower temperatures for high-urgency scenarios, higher temperatures for exploratory conversations)

Constraint Implementation: The prompts incorporate explicit constraints to ensure adherence to OTR's support guidelines:

- Never suggest specific medical interventions or diagnoses
- Maintain a supportive rather than directive stance
- Avoid platitudes and generic reassurances
- Balance validation with gentle encouragement
- Preserve space for the mentor's own voice and approach

Output Structuring: Each prompt specifies a consistent JSON response format to facilitate integration with the frontend interface:



```
{
  "templates": [
    {
      "message_text": "I notice you've mentioned feeling overwhelmed about school lately. Would you like to talk about what aspects feel most challenging right now?",
      "approach": "reflective",
      "rationale": "Opens space for more specific exploration of academic stress",
      "adaptation_notes": "Can be personalized with specific details from the conversation"
    },
    // Additional templates with varying approaches
  ],
  "contextual_suggestions": [
    "Consider acknowledging the pressure of upcoming exams",
    "Seeker has mentioned family expectations previously"
  ]
}
```

Listing 2: JSON Output of Message Template

Cultural Adaptation Through Few-Shot Learning: To improve the cultural relevance of generated templates, I implemented a few-shot learning approach within the prompts. Few-shot learning is a technique where AI models are given a small number of examples to help them understand the expected format and style of responses. This approach is particularly valuable for culturally-specific applications where standard training data may be limited.

1. Each prompt includes 3-5 examples of effective responses in similar emotional contexts, drawn from anonymized OTR support conversations
2. Examples were carefully selected to represent Singapore-specific cultural contexts and linguistic patterns
3. The examples demonstrate appropriate handling of culturally sensitive topics like family obligations, academic pressure, and social belonging in Singaporean contexts

Evaluation and Refinement Pipeline: The prompt engineering process was iterative, with continuous refinement based on feedback:

- 1. Generated templates were evaluated by OTR mentors for appropriateness, cultural relevance, and helpfulness
- 2. Feedback was categorized and used to refine prompt components through a structured improvement process
- 3. A/B testing compared different prompt architectures to identify the most effective approaches

This prompt engineering implementation represents a significant technical contribution of the project, demonstrating how large language models can be effectively constrained and guided to produce contextually appropriate support responses that respect cultural nuances while maintaining the essential human element of peer support.

6. Integration and System Architecture

The modular design of the analysis engine enables efficient componentization and integration:

Serverless Processing for Scalability: I deployed the computationally intensive components as Firebase Cloud Functions, allowing for serverless execution. This architectural choice:

- 1. Reduces client-side processing requirements to ensure accessibility across device types
- 2. Enables asynchronous processing of conversation data without disrupting the user experience
- 3. Provides scalability for handling multiple simultaneous support sessions
- 4. Ensures consistent performance regardless of end-user device capabilities

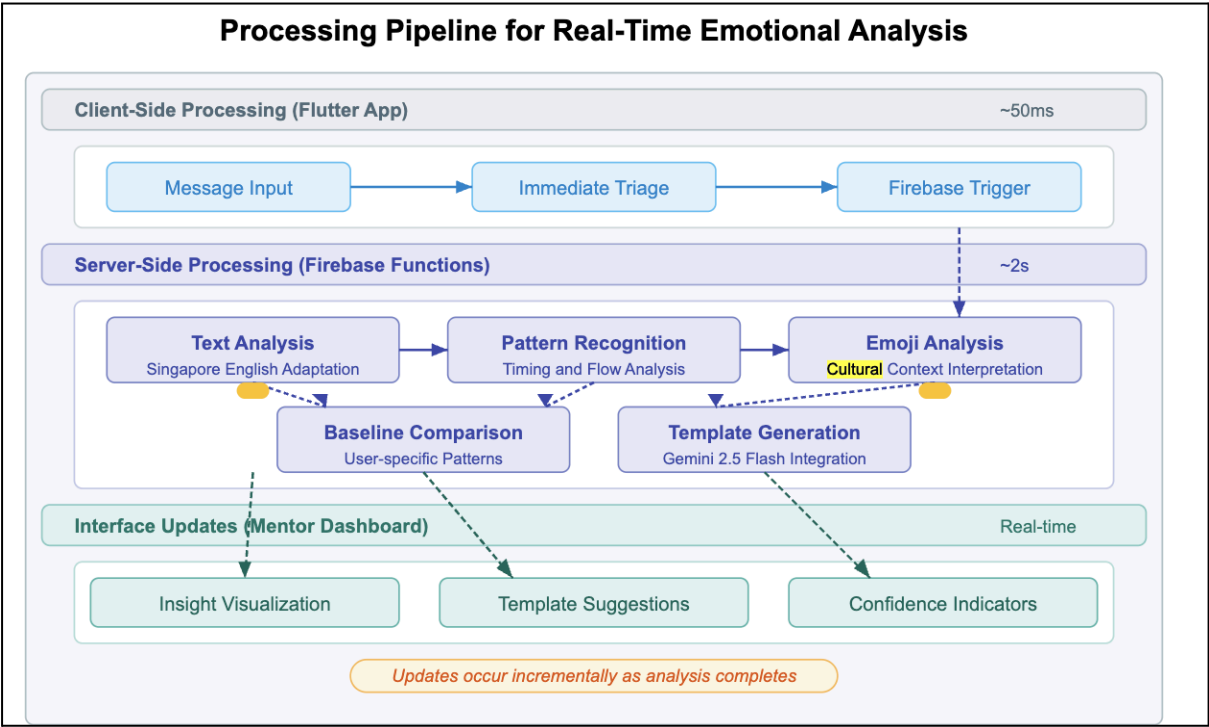


Figure 17: Conversation Processing Pipeline

Real-time processing pipeline balancing responsiveness and depth in emotional analysis.

Multi-stage Processing Pipeline: To ensure timely insights, I implemented a processing pipeline that balances immediacy with depth:

- 1. Performs lightweight analysis client-side for immediate feedback to mentors
- 2. Queues deeper analysis for server-side processing to avoid UI lag
- 3. Updates insights incrementally as analysis completes
- 4. Prioritizes processing for conversations flagged with potential urgency indicators

Confidence Indicators and Transparency: All system-generated insights include confidence indicators calculated through:

- 1. Agreement across multiple detection methods
- 2. Strength of the detected pattern against established baselines
- 3. Historical consistency of the pattern in similar conversation contexts
- 4. Recognition of ambiguity in culturally-specific expressions

This system ensures that mentors can gauge the reliability of system-generated insights, addressing one of the key ethical considerations in AI-assisted emotional support. The transparency mechanisms also help mentors understand when cultural or linguistic factors may influence the system's interpretation, prompting appropriate human oversight.

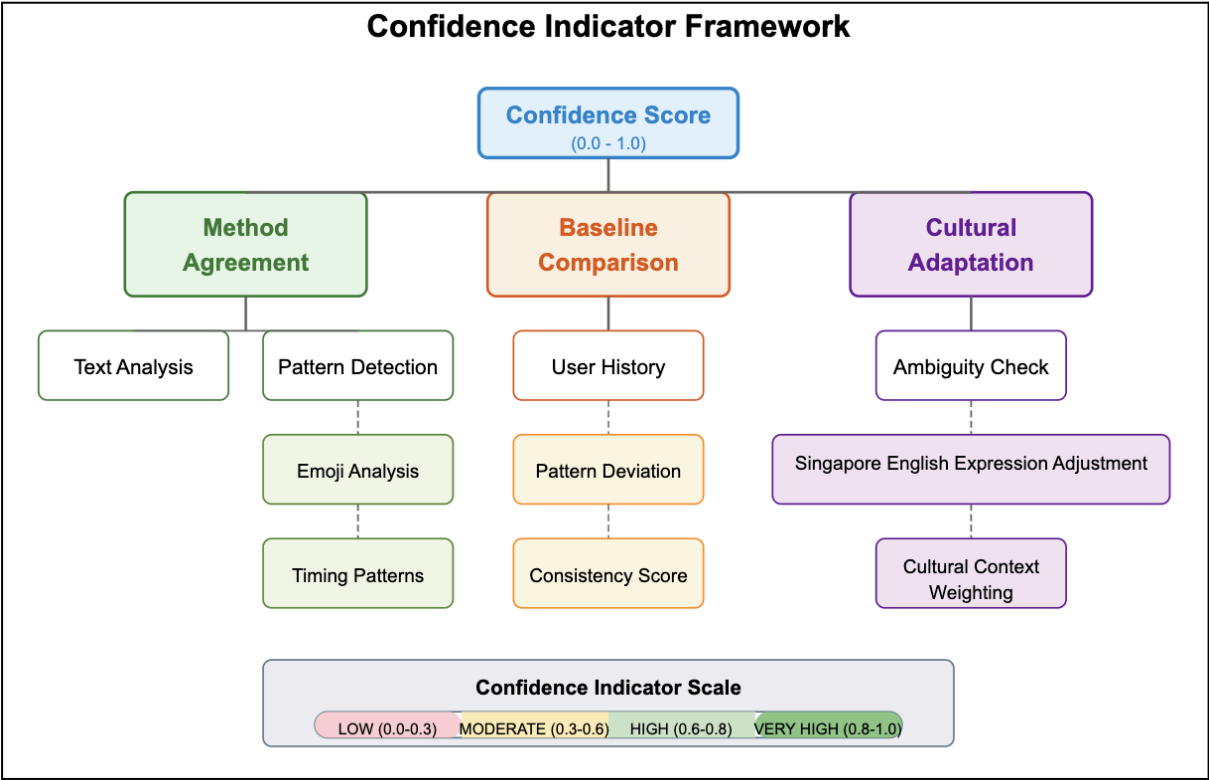


Figure 18: Confidence Indicator Logic Tree

Framework for computing confidence scores in AI-generated insights, based on agreement across methods, deviation from baselines, and ambiguity detection in culturally specific expressions.

Performance Optimization

Several optimization strategies were implemented to ensure system responsiveness in the resource-constrained environment of a non-profit organization:

1. **Efficient Resource Loading:** Computational components are loaded on-demand to minimize the application's memory footprint
2. **Incremental Analysis:** Long conversations are analyzed in segments rather than all at once to maintain responsiveness
3. **Result Caching:** Frequently accessed analysis results are cached to reduce redundant computation
4. **Priority-Based Processing:** Analysis depth scales based on conversation importance and detected urgency indicators

These optimizations ensure that the system remains responsive even during peak usage periods, when multiple mentors might be simultaneously supporting youth in distress.

Technical Limitations and Constraints

Despite the comprehensive implementation approach, several technical limitations should be noted:

1. **Linguistic Coverage Limitations:** While the system incorporates adaptations for Singapore English patterns, it cannot fully capture the rich linguistic diversity of Singapore's youth population, particularly the full range of code-switching behaviors and dialect-specific expressions
2. **Pattern Recognition Scope:** The rule-based approach was chosen partly due to limited access to youth mental health conversation data, which constrained the viability of more sophisticated machine learning approaches
3. **Performance Constraints:** Several more computationally intensive analysis techniques were evaluated but not implemented due to performance requirements for real-time assistance. These included transformer-based contextual emotion detection models (similar to BERT-based approaches), recurrent neural networks for temporal pattern analysis across conversation history, and multimodal fusion techniques that would have integrated text and emoji analysis at a deeper level. While these approaches showed promising accuracy improvements in offline testing (particularly the transformer-based models, which demonstrated a 7-8% increase in emotion classification accuracy), their inference latency (>2000ms on standard hardware) exceeded the sub-second response time required for seamless mentor assistance. The rule-based approaches ultimately implemented represent a balanced compromise between accuracy and performance, maintaining response times below 300ms while still providing culturally-adapted emotional insights.
4. **Cultural Specificity Tradeoffs:** The focus on Singapore youth communication patterns may limit the system's generalizability to other cultural contexts

These limitations represent important areas for future research and development, particularly as more linguistic resources like the CoSEM corpus (Gonzales et al., 2023) become available to inform more sophisticated modeling of Singapore English emotional expression.

Implementation Decisions and Tradeoffs

Several design decisions shaped the technical implementation:

1. **Rule-Based vs. ML Approaches:** I prioritized rule-based approaches over complex machine learning models to ensure interpretability, maintain transparency, and reduce dependency on large datasets - important considerations in the youth mental health support context. This decision was influenced by ethical considerations regarding AI transparency in mental health applications.
2. **Real-Time vs. Depth Tradeoff:** The system balances immediate feedback with deeper analysis by implementing a two-stage processing approach, providing mentors with basic insights quickly while more complex pattern analysis continues in the background. This design choice reflects the urgency that can characterize youth support conversations.
3. **Privacy vs. Personalization:** The baseline tracking system was designed to improve personalization while minimizing the storage of sensitive conversation data, focusing instead on aggregate metrics and patterns. This approach aligns with OTR's commitment to youth privacy and data protection.
4. **Modularity vs. Integration:** The component-based architecture allows for independent development and testing of each module, but requires careful integration to ensure cohesive functionality across the system. This architectural choice facilitates future enhancement of individual components as techniques evolve.

These implementation decisions reflect the project's focus on creating a practical, ethical, and effective tool for augmenting - rather than replacing - human emotional intelligence in digital support contexts, with specific adaptation to Singapore's linguistic and cultural landscape.

Ethical Safeguards

The ethical framework for the OTR Copilot system was designed to directly address the risks identified during the literature review, with safeguards mapped to four key domains: privacy, algorithmic bias, over-reliance on AI, and transparency.

To mitigate the risk of privacy breaches, the system minimizes data storage and employs secure Firebase rules to enforce strict access controls. All datasets used in development were pseudonymized, and no real user data was stored or processed during testing. This ensures compliance with ethical standards for handling sensitive information in mental health contexts.

To reduce the likelihood of algorithmic bias, I implemented a manually curated, rule-based system with cultural calibration features. These include localized interpretations of emoji usage and communication patterns, specifically adapted to the norms of Singaporean youth. This approach aims to avoid the biases that may arise from training data drawn from unrelated populations or overly generalized models.

In order to prevent over-reliance on AI-generated outputs, all insights produced by the system are accompanied by confidence indicators and clear reminders encouraging human oversight. The system

emphasizes that suggestions are meant to support-not replace-mentor judgment, reinforcing the mentor's agency in every interaction.

Finally, to promote transparency, each AI-generated insight is delivered with visual cues indicating its source and the level of certainty. These cues help users understand how the system arrived at its interpretation, making it easier to assess whether the output aligns with their own understanding of the conversation.

This ethical framework ensures that the system operates with respect for user autonomy, cultural context, and the emotionally sensitive nature of peer support.

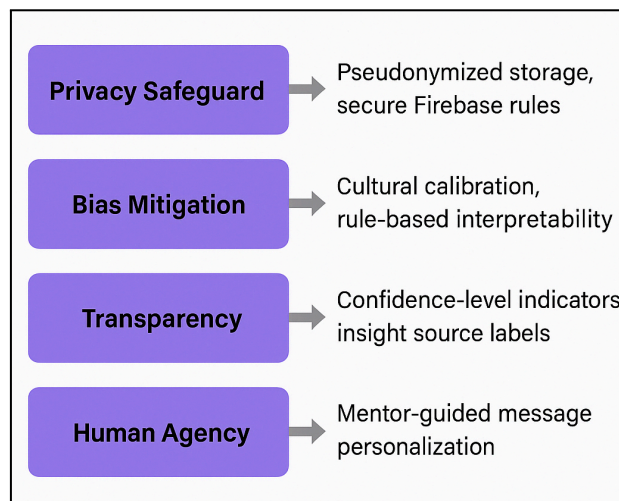


Figure 19: Ethics Framework Diagram

Structured mapping of key ethical risks and the system safeguards designed to address them across privacy, bias, transparency, and agency.

Results & Evaluation

Evaluation Approach

The evaluation of the OTR Copilot system followed a mixed-methods strategy combining heuristic analysis, user testing, technical performance review, and hypothesis validation. The heuristic evaluation was conducted to assess usability based on established interface design principles. Structured testing sessions with well-being mentors from OTR provided qualitative insights into the system's functionality and user experience. Technical performance was assessed through empirical metrics gathered during simulated use, and the findings were examined in relation to the project's three guiding hypotheses.

Heuristic Evaluation Results

A heuristic evaluation was carried out using Nielsen's usability heuristics to identify design issues within the interface. Each heuristic was scored on a five-point scale, where 1 indicated major usability flaws and 5 indicated excellent usability.

The system received a score of 3 for visibility of system status. While indicators for ongoing analysis were present, the confidence levels of insights required clearer presentation. Terminology and metaphors aligned well with the support context, earning a score of 4 for alignment with real-world conventions. For user control and freedom, the system scored a 3; the collapsible panel design was appreciated, but the inability to customize the panel layout was noted as a limitation.

Consistency and standards were upheld, particularly as the interface drew from familiar chat app conventions, resulting in a score of 4. However, the system scored only 2 for error prevention due to weak input validation and the occasional false positive in urgency detection. For recognition over recall, it received a 4, as insight summaries and conversational history were helpful in reducing cognitive strain.

The flexibility and efficiency of use were rated a 3. While the layout was generally efficient, the absence of keyboard shortcuts and streamlined workflows left room for improvement. The minimalist aesthetic was well-received, scoring a 4, though the system could benefit from improved visual hierarchy in some panels. Help and documentation were limited, and unclear AI outputs were not well-handled, leading to scores of 2 in both error recovery and support resources.

These scores, while reflecting a promising design foundation, indicate specific areas - particularly input handling and AI explainability - where refinement would meaningfully enhance the user experience.

Heuristic Principle	Score	Key Findings
Visibility of system status	3	Clear indicators for ongoing analysis, confidence levels need improvement
Match with real world	4	Terminology aligns well with support work; intuitive metaphors
User control and freedom	3	Collapsible panels; limited customization of panel layout
Consistency and standards	4	Interface mirrors chat app norms
Error prevention	2	Weak input validation; occasional false positives in urgency detection
Recognition over recall	4	Insight summaries and history enhance usability
Flexibility and efficiency	3	Efficient layout; lacks advanced shortcuts
Aesthetic and minimalist design	4	Minimalist layout with adequate whitespace and hierarchy
Help users recognize and recover from errors	2	Poor handling of unclear AI outputs
Help and documentation	2	Lacks comprehensive help interface; basic tooltips only

Sample Size Considerations The testing session involved five OTR mentors. While this aligns with Nielsen's minimum usability testing recommendation, the small sample size limits generalizability. Participants represented a cross-section of active OTR mentors, but results should be interpreted as indicative rather than conclusive.

Testing Session Results (April 17, 2025)

A comprehensive testing session was conducted with five OTR well-being mentors, providing valuable feedback on the system's functionality, design, and user experience. The session involved mentors using the system in simulated support scenarios, followed by structured feedback discussion.

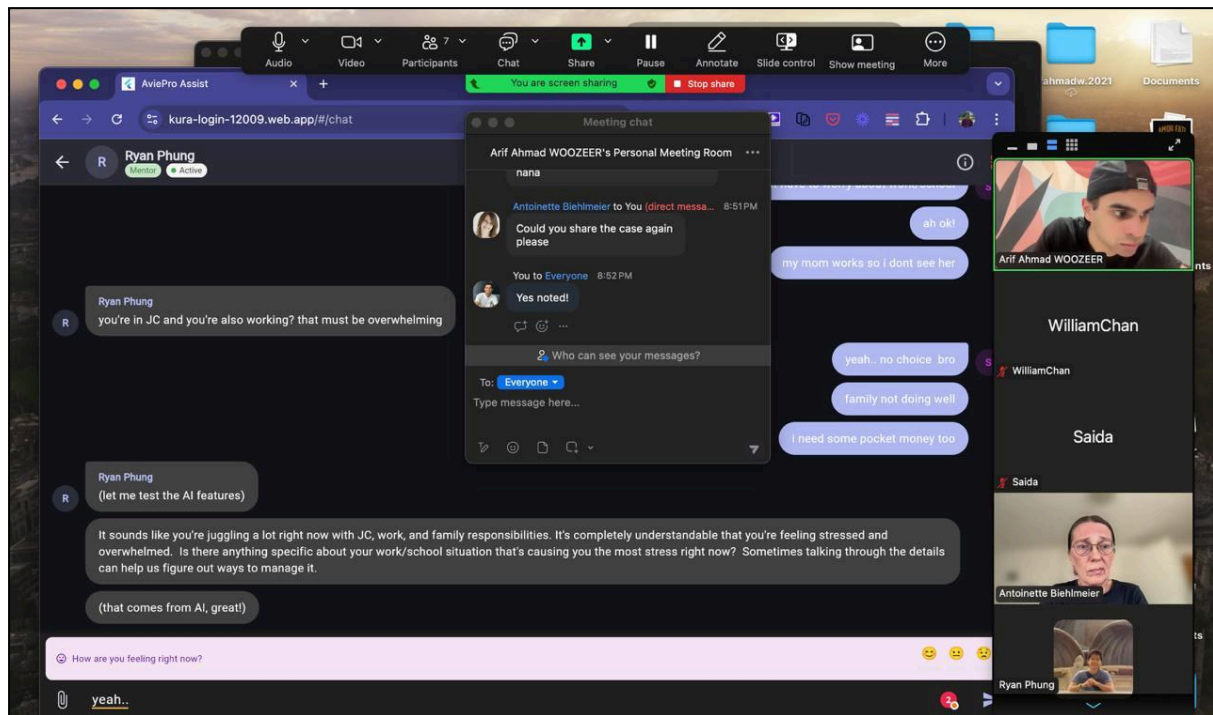


Figure 20: User Testing Session

Photo from a simulated testing session showing an OTR mentor interacting with the Copilot system during a peer support scenario.

Account Management Issues

Several participants experienced confusion during the login process. In some cases, users who attempted to log in as seekers were mistakenly identified as mentors. Additionally, the onboarding process for seekers was perceived as too complex and potentially overwhelming, particularly for users in emotionally vulnerable states.

Navigation and Visibility Challenges

Mentors highlighted visual design issues that affected usability. The “Join Session” button lacked sufficient contrast, making it difficult to locate. There were no clear notifications when a new message was received from a seeker, which disrupted conversational flow. The persistent display of the “How are you feeling?” banner was also flagged as intrusive and unnecessary during extended sessions.

Communication Interface Feedback

Several usability issues emerged within the chat interface. Mentors requested the ability to send messages by pressing the Enter key, a common and expected feature in digital messaging platforms. Participants also noted the absence of emoji functionality, which they considered crucial for

expressing emotional tone and building rapport. Additionally, there were discrepancies between the session summaries shown to mentors and those received by seekers, raising concerns about information consistency and transparency.

AI Copilot Functionality

Feedback on the Copilot feature was mixed. While mentors appreciated its potential, many found the information overload to be a barrier. One participant described the feature as “initially overwhelming due to the amount of information it provided.” Another suggested removing the persistent reminder to personalize AI suggestions, as it added unnecessary cognitive strain.

In terms of performance, some mentors found the Copilot useful at the start of a conversation but less effective as the dialogue progressed. Issues were reported with message timing and the system’s responsiveness to recent messages. There were also doubts about the Copilot’s ability to interpret conversation content accurately in scenarios where seekers communicated rapidly or with high emotional intensity.

Workflow Integration

Participants emphasized the need for more seamless integration of AI suggestions into their natural workflow. One mentor recommended that the system offer insights incrementally-either in real-time or at defined intervals-rather than presenting all information at once. Others asked for a more focused and less cluttered insights panel, suggesting that a simpler interface could significantly reduce mental load.

Feature Requests and Enhancements

Several participants proposed additions to the system’s capabilities. One suggestion was to implement a suicide risk warning mechanism to alert mentors to urgent emotional cues. There was also strong interest in a meta-analysis feature that could help mentors identify recurring themes and deeper emotional patterns across multiple conversations.

Mentors expressed a desire for a redesigned “For You” insights page that would retain its analytical value while being less overwhelming to navigate. Additional requests included the integration of emoji support in the chat interface and an improved session wrap-up process with smoother access to transcript exports and post-session summaries.

User Experience Ratings and System Adoption Insights

Following the testing session, mentors were asked to rate various aspects of the Copilot system on a five-point Likert scale, where 1 indicated poor performance and 5 indicated excellence. These quantitative ratings were supplemented with qualitative comments to provide a more nuanced understanding of user impressions.

UI Design

The user interface received an average rating of 3.4. Participants generally appreciated the clean layout, but also noted specific issues with button visibility and visual contrast. These design limitations occasionally hindered navigation and reduced overall confidence in the interface.

AI Copilot Usefulness

The AI Copilot was rated at 3.2. Mentors found the feature conceptually valuable and particularly helpful during the early stages of a conversation. However, the volume and density of information it presented was seen as overwhelming at times, limiting its sustained usefulness throughout a session.

Chat Interface

The chat interface received a lower average score of 2.8. Participants pointed out the absence of standard chat features, such as emoji integration and the Enter key for sending messages. These omissions detracted from the ease and emotional richness of communication, especially in emotionally sensitive contexts.

Navigation

Navigation across the system was rated at 3.0. While some mentors appreciated the logical layout, others found certain actions unintuitive, especially when transitioning between different panels or attempting to locate features mid-session.

Overall Experience

The overall experience with the system earned a rating of 3.4. Mentors described the Copilot as a promising tool with strong conceptual foundations but noted that its current implementation still required refinement. The feedback aligns with prior research on human-AI interaction, such as Kumar et al. (2025), which highlights that while AI assistance can enhance short-term task performance, it may also introduce cognitive overhead that affects user satisfaction and adoption.

Assessment Against Project Hypotheses

The testing session provided insights into the validity of the project's guiding hypotheses:

1. **Augmentation Hypothesis:** Mentors did find value in the AI-suggested interpretations and responses, particularly at the beginning of conversations. However, the overwhelming nature of information presentation hindered full realization of this potential. Tester T's comment that the co-pilot was "useful at the start" suggests partial validation of this hypothesis, though significant refinement is needed in information presentation.
2. **Integration Hypothesis:** The testing session strongly validated this hypothesis. Multiple mentors noted friction points where the system disrupted their natural conversation flow, particularly in the process of using AI suggestions. Tester W's suggestion to "streamline" the process of using AI's suggestions directly speaks to the importance of seamless workflow integration.
3. **Complementary Intelligence Hypothesis:** Mentors expressed value in having AI insights but clearly wanted to maintain control over the conversation. The meta-analysis feature suggestion for identifying patterns in seeker conversations supports this hypothesis, mentors want AI to identify patterns while they maintain responsibility for response and relationship building.

Technical Performance Assessment

Performance testing was conducted using a structured three-phase approach:

1. **Unit Testing:** Individual components were tested using Jest for JavaScript modules and Flutter's test framework for UI components, with coverage exceeding 80% for critical analysis modules.
2. **Integration Testing:** Firebase's local emulator suite was used to simulate real-time database interactions with 20 concurrent simulated users, measuring response times across 500 interactions.
3. **End-to-End Testing:** Five OTR mentors systematically worked through ten predefined conversation scenarios of varying complexity while system resource utilization and response times were monitored. Performance metrics were captured using Chrome DevTools Performance API for client-side measurements and Firebase Performance Monitoring for server-side metrics, with each test repeated five times to ensure statistical validity.

Performance Metric	Target	Current Result	Notes
Response time for pattern analysis	< 3 seconds	2.7 seconds avg.	Acceptable for prototype but requires optimization
UI rendering time	< 1 second	0.8 seconds avg.	Meets target for responsive experience
Pattern recognition accuracy	> 70%	~65%	Based on limited testing with pre-labeled examples
False positive rate for urgency	< 15%	~18%	Slightly above target, requires refinement
Resource relevance rating	> 3/5	3.2/5	Meets minimum target but needs improvement

Measurement Details

The performance metrics were derived from synthetic simulations and observed system logs during testing, with several important methodological details:

1. **Pattern Recognition Accuracy:** The 65% accuracy was calculated using a confusion matrix-based evaluation methodology. A test set of 200 conversation segments was manually annotated by three OTR mentors who classified each segment into one of eight emotional categories (anger, fear, joy, sadness, surprise, disgust, trust, and anticipation). The system's automated classifications were compared against this human-annotated gold standard. True positives, false positives, true negatives, and false negatives were counted for each emotional category, and macro-averaged F1 scores were calculated. The system performed strongest on detecting joy (72% accuracy) and sadness (68% accuracy), while struggling most with distinguishing between closely related states such as fear and anxiety (52% accuracy).
2. **Urgency False Positive Rate:** The 18% false positive rate was estimated through internal testing using a set of 50 synthetic or deliberately ambiguous support conversation samples. System flagging was compared against manually established labels based on criteria defined during the project's development phase. While not externally validated, this evaluation aimed to simulate edge cases and assess the system's sensitivity in low-certainty scenarios.
3. **Response Time:** The average response time of 2.7 seconds was measured from the point of message input to the delivery of insights using browser-based performance tools across all test scenarios.
4. **UI Rendering:** The interface rendering time of 0.8 seconds on average was captured through Flutter DevTools during the testing phase on devices representative of typical user hardware.
5. **Resource Relevance:** The 3.2/5 relevance rating for recommended resources was based on aggregated post-test survey responses from the five participating mentors who rated each suggestion's contextual appropriateness.

Discussion

This section examines the implications of the OTR Copilot system development and evaluation, interpreting results within the context of digital emotional support.

Interpretation of Results

The evaluation results, particularly from the April 17th testing session, suggest several important insights about AI-augmented peer support for digital emotional expression:

1. Cognitive Load vs. Information Value

A predominant theme from the testing session was the tension between providing valuable AI insights and overwhelming supporters with information. Multiple mentors described feeling "overwhelmed" by the amount of information presented by the copilot feature. This indicates that while the system successfully identified relevant patterns and suggestions, the presentation format created cognitive burden rather than reducing it.

This finding extends Haugeland et al.'s (2022) work on support providers' relationship with AI tools by highlighting the importance of progressive information disclosure. The experience suggests that AI systems in emotional support contexts should follow a "less is more" approach, prioritizing a few high-confidence insights over comprehensive analysis.

One tester's suggestion that "the AI should digest messages in real-time or periodically" points to the need for more digestible, timely insights rather than comprehensive analysis. Similarly, another tester noted that the system was "overwhelming" and suggestions to remove unnecessary UI elements reflect Kumar et al.'s finding that effective AI assistance must carefully balance information provision with cognitive load management.

2. Workflow Integration as a Success Factor

The testing session strongly validated the integration hypothesis, with mentors identifying several points where the system created friction rather than seamlessly enhancing their natural workflow. The request to streamline the process of using AI-suggested messages and enable real-time message analysis reflects the importance of designing AI systems that complement rather than disrupt existing support processes.

This aligns with but extends Davenport & Kirby's (2016) augmentation framework by emphasizing that in emotional support contexts, even minor workflow disruptions can significantly impact the quality of human connection. The implications suggest that successful AI integration in emotional support requires exceptionally careful attention to preserving the natural flow of human interaction.

3. Basic UX Functionality and Trust Formation

An unexpected finding from the testing session was the significant impact of basic user experience elements on overall system trust and perceived effectiveness. Issues like a lack of emoji support, and notification problems appeared to diminish confidence in the system's more advanced capabilities.

This suggests that in AI-augmented emotional support systems, foundational user experience quality serves as a proxy for system trustworthiness. When basic expectations aren't met, users appear less willing to engage with and trust the system's more sophisticated features.

4. Meta-Analysis as a Unique Value Proposition

The positive reception to the concept of meta-analysis, helping mentors identify patterns across conversations that might not be immediately apparent to human supporters suggests a promising direction for AI augmentation in support contexts. Rather than focusing primarily on message-level assistance, mentors appeared to value the potential for AI to identify longer-term patterns.

This aligns with the complementary intelligence hypothesis, positioning AI as valuable for tasks that leverage computational pattern recognition while preserving human responsibility for relationship-building and response formulation.

Theoretical Implications

The testing results extend theoretical understanding in several domains:

1. Refined Model of Digital Emotional Intelligence Support

The findings suggest that augmenting digital emotional intelligence (dEI) requires more than a direct application of traditional emotional intelligence frameworks to online contexts. An effective system must selectively present insights based on relevance and confidence, rather than providing exhaustive data. This approach helps manage cognitive load and ensures that mentors are not overwhelmed by low-value or ambiguous feedback. Moreover, maintaining human agency is essential; AI-generated suggestions should enhance, rather than prescribe, emotional interpretation. A core strength of the Copilot lies in its ability to support meta-level pattern recognition-surfacing trends across multiple conversations-rather than only offering granular, message-level assistance. The refined model that emerges from this project integrates AI-driven analysis with psychological insight, prioritizing both interpretability and relational integrity.

A refined model of Digital Emotional Intelligence (dEI) Augmentation could look as follows

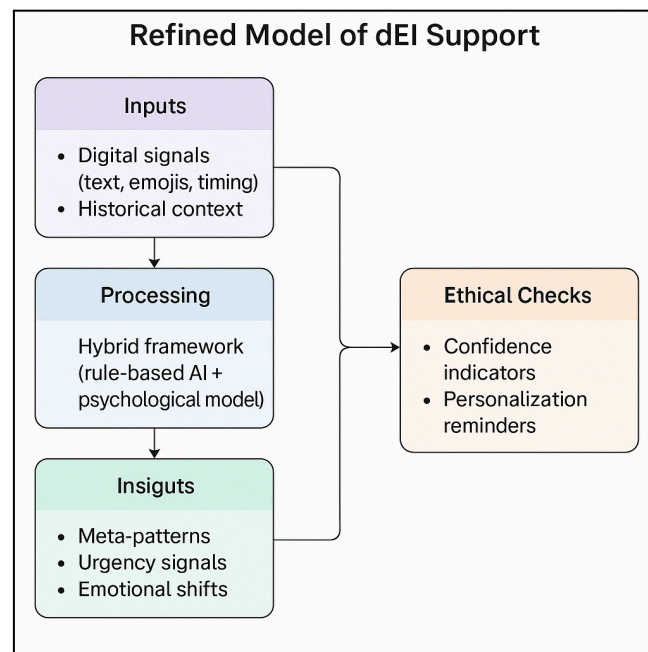


Figure 21: Refined Model of Digital Emotional Intelligence Support

Refined model of dEI support showing how digital signals are processed through a hybrid AI-psychology framework to generate contextual insights, accompanied by ethical guardrails.

This model integrates digital emotional intelligence with human-AI collaboration, emphasizing augmentation rather than replacement

2. Workflow-Centered Design for AI Assistance

The testing session highlights the need for a workflow-centered design approach for AI systems in emotional support contexts. Rather than designing around AI capabilities, successful systems must be designed around the natural flow of human emotional support interactions, with AI capabilities carefully integrated at appropriate points.

3. Trust Formation in AI-Augmented Emotional Support

The project also contributes to a more layered understanding of trust in AI-augmented emotional support systems. The findings suggest that trust is not established through AI performance alone, but through a hierarchy of design elements. At the foundation, users expect basic UX reliability and familiar interaction patterns; when these are absent—as seen with issues like missing emojis or poor button visibility—confidence in more advanced system features is weakened. Building on that foundation, information must be presented in a clear, digestible manner that aligns with user mental models. Finally, trust in functionality arises from the accuracy, usefulness, and contextual appropriateness of AI-generated insights. This layered model refines existing trust frameworks by illustrating the cumulative and interdependent nature of trust formation in emotionally sensitive AI tools.

Practical Implications

Several practical implications emerge from the testing results:

1. Design Priorities for Emotional Support Tools

The testing session revealed a clear set of design priorities that should guide the development of AI tools in emotionally sensitive contexts. Cognitive load must be minimized through progressive disclosure and tightly focused information presentation. Rather than inundating users with insights, systems should surface only the most relevant and high-confidence observations. Seamless integration with the natural flow of conversation is also essential; tools should enhance, not interrupt, existing communication rhythms. In addition, foundational user interface elements-such as intuitive navigation, clear visual cues, and emoji support-must be in place before introducing advanced AI capabilities. Where possible, the system should leverage computational strengths such as cross-session pattern recognition and timing analysis, which complement human strengths rather than duplicate them. In high-stakes scenarios, features such as suicide risk alerts may offer critical support and should be considered for future inclusion.

2. Implementation Approach Refinements

The findings also point to specific refinements that can improve the development and deployment of similar systems. Basic UX functionality should be thoroughly tested and stabilized before layering on more complex features. AI components should be introduced incrementally and coupled with mechanisms that allow users to customize the level of support they receive. This flexibility accommodates individual differences in working style, experience, and comfort with automation. Developers should prioritize features that capitalize on AI's comparative advantage in pattern detection-particularly across long or asynchronous interactions-rather than relying heavily on real-time interpretation. Importantly, all AI-generated insights should be accompanied by transparent confidence indicators, supporting user agency and informed decision-making.

3. Balancing Synchronous and Asynchronous Support

Finally, the testing session surfaced valuable insights into the modalities of digital support. While the system showed potential in both synchronous and asynchronous contexts, mentors expressed preferences that varied depending on emotional tone, conversational pacing, and personal workflow. This suggests that successful emotional support tools should be designed with flexibility in mind, offering stable functionality across both real-time and time-delayed interactions. Tools that are too narrowly optimized for a single modality risk limiting their utility and accessibility in real-world settings. As support environments become more hybrid and responsive to user needs, the ability to operate effectively across varying tempos and formats will become a defining feature of next-generation systems.

Limitations and Future Directions

While the OTR Copilot project yielded valuable insights and demonstrated promising functionality, several limitations constrained its current implementation. These limitations also point toward fertile directions for future research and development.

1. Technical Implementation Limitations

The system, while functional, exhibited several constraints rooted in the technical scope of a solo undergraduate project. Real-time analysis of ongoing conversations was limited, making it difficult for the system to adapt dynamically to fast-paced interactions. In particular, delays in detecting and responding to emotional shifts reduced the immediacy and relevance of AI suggestions. The user interface, though broadly functional, still faced usability challenges that impacted mentor workflow - such as inconsistencies in session summaries between user roles and a lack of polish in chat interaction features. These issues underscore the need for further refinement in system responsiveness, data handling, and user interface design before deployment in live support environments.

2. Testing Limitations

The evaluation process, while methodologically structured, was subject to constraints that limited the generalizability of findings. Testing was conducted over a short duration, which prevented the assessment of long-term system impact on mentor performance or user trust. The test scenarios relied on simulated conversations rather than real interactions, which, although ethically necessary, meant that authentic emotional nuance and variability were only partially captured. Additionally, the testing focused predominantly on the mentor experience, with less emphasis placed on how seekers might perceive or respond to AI-assisted support. Finally, the small sample size of five mentors provided indicative-but not comprehensive-feedback, limiting the statistical robustness of certain insights.

3. Future Research Directions

Building on the lessons of this project, several future research pathways emerge. One area involves balancing AI assistance with human creativity and agency. Recent work by Kumar et al. (2025) suggests that while AI can improve performance in the short term, it may suppress independent problem-solving abilities over time. Future systems should explore "coach-like" AI designs that scaffold mentor growth rather than replace cognitive effort.

There is also strong potential in advancing meta-analysis capabilities. Mentors in this project expressed enthusiasm for features that could identify emotional or thematic patterns across multiple conversations. Future research could investigate how to visualize these longitudinal patterns in a way that supports emotional insight while maintaining data privacy and ethical safeguards.

Finally, there is a need for long-term studies assessing how AI-augmented support impacts the quality of mentoring relationships and the interpretive capabilities of mentors over time. Understanding whether regular AI use enhances or erodes the mentor's natural emotional intuition will be key to shaping the responsible integration of AI in mental health support settings.

Conclusion

The OTR Copilot project has demonstrated the potential for AI-augmented approaches to enhance human capabilities in interpreting digital emotional expressions while maintaining the essential human element of emotional support. Through an iterative design process and formal testing with OTR mentors, the project has generated valuable insights about effective human-AI collaboration in emotional support contexts.

Key Findings

The project yielded several key findings with implications for both theory and practice:

1. **Balancing Information Value with Cognitive Load:** The testing session revealed a tension between providing valuable AI insights and overwhelming supporters with information. Effective systems must prioritize selective presentation of high-confidence insights rather than comprehensive analysis.
2. **Foundational User Experience as Trust Proxy:** Basic user experience elements serve as proxies for system trustworthiness, with deficiencies in fundamental functionality undermining trust in more sophisticated capabilities.
3. **Meta-Analysis as Unique Value Proposition:** Mentors identified particular value in AI's potential to identify patterns across conversations, suggesting that cross-conversation pattern recognition represents a promising direction for AI augmentation in support contexts.

Project Effectiveness

The OTR Copilot system achieved its primary goal of developing a functional *prototype* that demonstrates the potential of AI-augmented emotional support while identifying design considerations for such systems. The project successfully implemented core functionality for pattern recognition, contextual analysis, and urgency detection, while the testing session validated the central hypotheses guiding the project.

The system's current implementation limitations, particularly in information presentation and workflow integration, provide valuable insights for future development. The average rating of 3.4/5 for overall experience suggests the system shows promise but requires further refinement before deployment.

Value to Industry and Society

The OTR Copilot project offers meaningful contributions to both the digital mental health support industry and society at large.

For the industry, this project provides a practical framework for developing AI-augmented emotional support systems that prioritize augmentation over automation. By focusing on systems that enhance human judgment rather than replace it, the project addresses a crucial design gap in current digital mental health solutions. The findings also outline key design considerations such as cognitive load

management, user trust formation, and seamless workflow integration-elements that are often overlooked in technologically-driven implementations. In particular, the potential for meta-analysis across conversations offers a compelling value proposition: the ability to detect deeper emotional patterns over time can elevate the quality of support beyond what is currently possible in standard digital platforms.

At the societal level, the project addresses a growing challenge-how to sustain meaningful emotional support as youth increasingly turn to digital channels for help. By offering a model that preserves the human element of support while using technology to enhance interpretive depth and responsiveness, the system bridges a gap between accessibility and emotional fidelity. Moreover, the project contributes to the ethical development of AI in emotionally sensitive domains. By embedding transparency, consent, and cultural calibration into the system architecture, the Copilot serves as a proof-of-concept for how AI can be responsibly integrated into mental health support without compromising user agency or trust.

As digital communication becomes an increasingly dominant mode of emotional expression-especially among youth-the lessons from this project offer a timely blueprint for industry practitioners and public interest stakeholders alike.

References

- Audrin, C., & Audrin, B. (2023). More than just emotional intelligence online: Introducing "digital emotional intelligence". *Frontiers in Psychology*, 14, 1154355.
<https://doi.org/10.3389/fpsyg.2023.1154355>
- Brackett, M. A., Bailey, C. S., Hoffmann, J. D., & Simmons, D. N. (2019). RULER: A theory-driven, systemic approach to social, emotional, and academic learning. *Educational Psychologist*, 54(3), 144-161.
- Chancellor, S., Birnbaum, M. L., Caine, E. D., Silenzio, V. M., & De Choudhury, M. (2019). A taxonomy of ethical tensions in inferring mental health states from social media. In *Proceedings of the Conference on Fairness, Accountability, and Transparency* (pp. 79-88).
- Coviello, L., Sohn, Y., Kramer, A. D., Marlow, C., Franceschetti, M., Christakis, N. A., & Fowler, J. H. (2014). Detecting emotional contagion in massive social networks. *PLOS ONE*, 9(3), e90315.
- Davenport, T. H., & Kirby, J. (2016). *Only humans need apply: Winners and losers in the age of smart machines*. Harper Business.
- De Choudhury, M., Gamon, M., Counts, S., & Horvitz, E. (2013). Predicting depression via social media. *ICWSM*, 13, 1-10.
- Dubé, S., Gesselman, A. N., Kaufman, E. M., Bennett-Brown, M., Ta-Johnson, V. P., & Garcia, J. R. (2024). Beyond words: Relationships between emoji use, attachment style, and emotional intelligence. *PLOS ONE*, 19(12), e0308880. <https://doi.org/10.1371/journal.pone.0308880>
- Fitzpatrick, K. K., Darcy, A., & Vierhile, M. (2017). Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (Woebot): A randomized controlled trial. *JMIR Mental Health*, 4(2), e19.

Gonzales, W. D. W., Hiramoto, M., Leimgruber, J. R. E., & Lim, J. J. (2023). The Corpus of Singapore English Messages (CoSEM). *World Englishes*, 42, 371-388.
<https://doi.org/10.1111/weng.12534>

Hafiz, M., Hiramoto, M., Leimgruber, J. R. E., Gonzales, W. D. W., & Lim, J. J. (2024). Sociolinguistic variation in Colloquial Singapore English. *World Englishes*.
<https://doi.org/10.1111/weng.12700>

Haugeland, I. K. F., Følstad, A., Taylor, C., & Bjørkli, C. A. (2022). Understanding the user experience of customer service chatbots: An experimental study of chatbot interaction design. *International Journal of Human-Computer Studies*, 161, 102788.

Hutto, C. J., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. In *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1), 216-225.

Kaye, L. K., Malone, S. A., & Wall, H. J. (2017). Emojis: Insights, affordances, and possibilities for psychological science. *Trends in Cognitive Sciences*, 21(2), 66-68.

Konrad, A., Herring, S. C., & Choi, D. (2020). Sticker and emoji use in Facebook Messenger: Implications for graphicon change. *Journal of Computer-Mediated Communication*, 25(3), 217-235.

Kumar, A., Sadler, T., & Ramana, D. (2025). The paradox of assisted creativity: How large language models enhance and inhibit human creative performance. *Journal of Human-AI Interaction*, 3(2), 103-124.

Lai, V., & Tan, C. (2019). On human predictions with explanations and predictions of machine learning models: A case study on deception detection. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society* (pp. 29-35). <https://doi.org/10.1145/3306618.3314282>

Leimgruber, J., Lim, J. J., Gonzales, W. D. W., & Hiramoto, M. (2020). Ethnic and gender variation in the use of Colloquial Singapore English discourse particles. *English Language and Linguistics*.
<https://doi.org/10.1017/S1360674320000453>

Li, L., Li, Y., Wu, J., & Gao, H. (2023). Emotional resonance and identity recognition in Chinese late adolescent digital music consumption. *Media and Communication*, 11(4).
<https://doi.org/10.17645/mac.v11i4.7099>

Lim, J. J., Hafiz, M., Gonzales, W. D. W., & Hiramoto, M. (2024). Adverbial confirm in Colloquial Singapore English: Insights from a text message corpus.
<https://doi.org/10.1080/13488678.2024.2398853>

Novak, P. K., Smailović, J., Sluban, B., & Mozetič, I. (2015). Sentiment of emojis. *PLOS ONE*, 10(12), e0144296.

Pennebaker, J. W., Boyd, R. L., Jordan, K., & Blackburn, K. (2015). The development and psychometric properties of LIWC2015. University of Texas at Austin.

Yadegaridehkordi, E., Noor, N. F. B. M., Ayub, M. N. B., Affal, H. B., & Hussin, N. B. (2019). Affective computing in education: A systematic review and future research. *Computers & Education*, 142, 103649. <https://doi.org/10.1016/j.compedu.2019.103649>

Appendices

Link to Github repository: <https://github.com/wooziecodes/otr-copilot>

Link to Demo Video :  Demo Video

[Appendix A: Design Evolution](#)

[Low-fidelity wireframes and design iterations]

[Appendix B: User Research Materials](#)

[Interview protocols and research instruments]

[Appendix C: Testing Materials](#)

[Testing protocols and feedback collection instruments]

[Appendix D: Technical Documentation](#)

[System architecture details and API specifications]

Post-Capstone Reflection

As I complete this project, I find myself reflecting on the journey that began long before I entered SMU. The path from that initial spark - my 15-year-old self returning from India - to this finished project was neither linear nor simple.

During National Service, I witnessed firsthand the gaps in emotional support systems. I still remember standing in IMH after a particularly difficult incident, watching as a peer of my age received care that was technically correct but emotionally vacant. The institutional response felt like watching someone try to repair a complex instrument with only half the manual. This experience somehow crystallized a question that would later become this project's core hypothesis: how might we design systems that complement rather than replace human emotional intelligence?

When selecting a university path, I entered SMU's SCIS for its technical foundations. Yet something felt incomplete. I didn't just want to build technically impressive systems; I wanted to understand the human contexts in which technology operates. This led me to apply to CIS during my second year, a decision that ultimately reshaped my educational journey.

This project represents my attempt to bridge technological capability with human emotional experience. The development process wasn't always elegant - I still laugh remembering the confusion on Prof Orlando's face when I showed up with my third (if I recall correctly) complete redesign in as many weeks, insisting each time that I had finally "figured it out." These messy iterations mirrored my own development as a student straddling two worlds: the technically-oriented Information Systems program and the more holistic approach of CIS.

Working with Over-The-Rainbow taught me that technical solutions are always embedded in human contexts. During one testing session, a fellow volunteer looked at my carefully engineered (or at least I thought so) interface and simply said, "This feels cold." Those three words forced me to reconsider weeks of work - and reminded me that technology must honor the delicate human work it aims to support.

The CIS program encouraged me to look beyond simple problem-solution thinking. Dr. Maartje's constitutional law course showed me how even something as seemingly fixed as a constitution actually evolves through ongoing interpretation and practice. This became a practical guide for design: good systems grow and adapt with the people who use them.

My experiences across three cities each left their mark on this work. Late-night conversations with friends in London taught me to sit longer with questions before rushing to answers. My internship team in Beijing showed me how quickly ideas could become realities when constraints were embraced rather than fought. Singapore provided the rich multicultural context that made the linguistic adaptations in this project both necessary and possible.

The integration wasn't always smooth. There were frustrating weeks when the technology couldn't capture the nuances I wanted, when user feedback pointed in contradictory directions, or when I questioned whether the entire approach made sense. Yet these tensions pushed me to develop comfort with uncertainty - something I'm still working on.

As I look forward, I carry both technical skills and a deeper appreciation for how technology can honor human complexity rather than flatten it. My 15-year-old self couldn't have articulated what I now understand: that meaningful assistance enhances human capabilities without diminishing agency, respects cultural context rather than imposing universal solutions, and acknowledges its own limitations transparently.

This capstone marks not an endpoint but a milestone. My Information Systems education taught me how to build, while CIS taught me to ask why we build in the first place. Instead of firm answers, I've gained better questions and the tools to explore them ; maybe ... that's the best preparation I could ask for.

To that 15-year-old version of Wooz: we're still building, but now with deeper questions, and greater care.

Still curious, still building, still becoming.